

Universe24: one integer rule with the cube's 24 rotations gives the lattice Klein-Gordon and Poisson equations, with Schrödinger's as its slow limit

Alon Gonen

Independent researcher, Haifa, Israel

alon@defounder.ai ORCID 0009-0008-2701-4599

October 2026

Abstract

We present one method of computing physics: a GameBoard of Nodes in whole numbers, read through clicks, on which the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, with nothing continuous beneath; every formula below is an example of its yield, Section 10.3 listing what it does not yet give. Rule3 steps every Node from its present, past and six neighbours by one division, reversibly, conserving a quadratic form, covariant under the translations and the cube's 48. Measurement is the meeting of the record run forward with the one read backward. Schrödinger's equation is its slow limit, light's index in a well e^{2U} . Newton's gravitation is derived: Poisson's equation a field's rest, the pull a packet's fall, G a reading. Of Einstein the weak field is derived in the clock's potential under one named assumption, the exponential metric in the lattice's frame, a Link clocked twice by its Node, $\beta = \gamma = 1$ exactly, a shadow 4.6 percent above Schwarzschild's, and the relativistic part only as the gap closes; the structure is a scalar pace, not the tensor equations, and against Kepler's potential the computed pair departs by 25 to 33 percent, a gap below 10^{-18} . The three experiments are derivations, not runs: the two-slit row is the line stepped on its world, and Bell's $S = 2\sqrt{2}$ and GHZ's $M = -4$ follow from the line's linearity through the pair's root under one declared credit, every local credit giving at most 2.

Keywords: integer lattice rule; reversible cellular automaton; measurement as a meeting of past and future; Bell's inequality; weak-field gravity; limiting speed on a lattice

1 Introduction

The question. The object of this paper is one method of computing physics: a GameBoard of Nodes stepped by one local, reversible rule in whole numbers, read only through clicks, the one measurement there is. On it the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, Rule3: the line itself is the lattice wave equation, its count read in quanta is the click, and its paces read as a clock and a ruler are the metric, with nothing continuous beneath. The question is whether that method can give the clicks that nature shows, and with how little state, non-locality and randomness; the fifteen formulas and the three experiments below are examples of what the method yields, not its scope, and Section 10.3 lists what it does not yet give. Of Newton the paper derives gravitation whole: Poisson's equation as a field's rest on the lattice, the inverse-square pull as a free packet's fall in the field's gradient at the band's inertia, the mass as the count, Kepler's orbits, and G as a reading of the law's constants. Of Einstein it derives the weak field in the clock's potential, the redshift, the bending, Shapiro's delay, the perihelion and the geodetic precession, with $\gamma = \beta = 1$ exactly under one named assumption, a Link clocked twice by its reading Node's own clock, and the special-relativistic part only as the matter family's gap closes, the moving clock's factor reaching Lorentz's as the family's limiting speed reaches light's; the structure is a scalar pace and not the tensor field equations, so it departs from Einstein's at the second order and in the strong field, has no frame dragging and not the tensor's polarisations, and against the Newtonian potential the orbits measure the computed pair's departures are the band's gap, closing with it (Section 8). It starts from one statement and never contradicts it (ASSUMPTION): the world is the meeting of the past with the future; a click is that meeting, the one measurement there is; and an observer sees only clicks (the words are defined at the end of this section). In that sense the model is the least computation we know of that gives the two slits and Bell's correlation under the declared credit; of nature's other clicks it gives the forms of Section 8 at the computed pair, with the findings of Section 10.3 against them. A Node holds a record's present and its past, two integer levels, and

the rule makes the record's future from the two and from the six neighbouring Nodes. The rule is covariant under the lattice's translations and under the symmetry group of the cube, the 24 rotations and, with the reflections, the 48 signed permutations of the three axes; no boost is among them (Section 3.1). The lattice holds nothing but events that split: at every Node a level splits by the rule to the six neighbours. Run forward from an emitter, the split events are the future, every path at once; read backward from a realised click by the same rule, they are the past, and through two slits both pass both gaps. A click is where the two meet at a declared detector, and which meeting is realised is the one thing the model does not determine, the one non-local element of the description (Section 5.2). What appears as a wave is the future, what appears as a particle is the present, the click, and the track is the past (Section 5.1). Least is meant in three named counts, the state per Node, the non-local elements and the random draws, among the models we know of; no theorem of minimality is claimed, and why nature is so we do not claim to know.

The rule in one line. Everything in the paper follows from one line of integer arithmetic, Rule3, together with one write and three readings. At every Node, in every interval and for every family of physical values [1, the line; the booking; the primitives]:

Rule3:	$w a_{\text{next}} + r' = \sum_{a \in \{x, y, z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2);$	
the one write:	a held part at a Node gains $\left(\sum_f w_f q_f(\text{Node}) + r \right) \text{div } E$, the remainder r kept there;	(1)
the readings:	$D_i = \text{now}_i^2 - \text{next}_i \text{before}_i, \quad W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j).$	

In the first line a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval later: the present, the past and the future. $\text{arr}_{\pm a}$ is the arrival through the Port $\pm a$, the neighbour's level. r and r' are the remainder kept at the Node before and after the step, Γ the Node clock. $[\text{num}, \text{den}]$ is the family's pair, and $\cos \omega_0 = \text{num}/\text{den}$ is its rotation per interval at wave number zero. p_0 is the Node's clock pace and p_a its pace along the axis a ; both are the Node clock lowered by the field levels the family reads there (Section 3.4). div is the floor division. The second line is the only way a field's level is ever written: the sum over the sourcing families f of each family's own quantity q_f at the Node, weighted by w_f , plus the part's one remainder, divided by the part's wall E . The third line is what is read from a record and never written: its form D_i , which sources the fields; its Wronskian W_i , its charge; and its current F_{ij} into the Node i from its neighbour j , which a detector reads through its boundary in quanta of $W_c = 3 \text{ den } T$, with T the quantum's norm, the normalisation of the quadratic form.

The formulas derived. From Eq. (1), in the paper's order, each with its mark in Table 1 and its algebra in the supplement: light's band and dispersion (Section 3.3); light's index e^{2U} in a well (Section 3.4); the lattice Klein-Gordon equation and its Schrödinger limit with the inertia $3 \tan \omega_0$ (Section 3.3); the conserved form and the share identity (Sections 4.3 and 4.4); the Wronskian as the charge (Section 5.5); Poisson's and Yukawa's rests, $\cosh \kappa = 3 \text{ den}/\text{num} - 2$ (Section 6.4); the bound body's functional and its stability (Section 7.2); the adiabatic invariant $A^2 \sin \omega$ and the count's drift (Section 7.4); the clock's rate $f(\omega_0)U$, the bending $4U$, Shapiro's delay, the perihelion and the shadow's capture radius $2em$ (Section 8); the moving clock, Lorentz's factor reached as the family's limiting speed reaches light's (Section 8, (e)); de Broglie's and Planck's relations and $E = mc^2$ as $m^* c_m^2 = \omega_0$ (Sections 8, (h), and 3.3); the content's index for every family, light's e^{2U} and matter's growing as $1/k^2$, the lens and the fall as refraction (Section 8, (p)); Bell's E and $S = 478/169$, Tsirelson's bound and GHZ's $M = -4$ under the declared credit (Section 9.4); the fine-structure constant $(3\sqrt{3}/8\pi)/(\Gamma \sin \omega_0)$, the units and the highest photon energy (Section 10.1); and Newton's G as a reading (Section 10.1). Fifteen formulas from five fields, every one an algebraic consequence of the line and of the inputs named beside it.

What the paper contributes. We state the rule and prove what is exact about it, reversibility to the bit, the conserved form with its current, the momentum flux as the wave's own stress (Section 4); we formulate measurement as the meeting of the forward and the backward record at a declared detector, the one non-local element named (Section 5); we derive the families as bands of the rule and choose a universe with as few free numbers as the rule allows (Section 6); we show that bodies are fixed points, prove their stability under the dilation and state how they cluster (Section 7); we derive the weak-field forms of gravity and the moving clock's factor, each marked by what it rests on, in the model's own potential and in the one nature's tests read (Section 8); and we state the two slits and Bell as derivations of what each must give, neither experiment performed here (Section 9).

What is put in, and the three kinds of claim. The paper’s claims are of three kinds, and the text keeps them apart. *Found*: what is proved or derived from the rule with no input chosen for it, the fifteen formulas above and the click formulas of Table 2 marked derived; the findings against nature are comparisons, listed in Section 10.3. *Assumed, chosen or declared*: the assumption and the credit below, the universe’s rows, the detector’s four declarations, its click’s whole among them, the electron as the free quantum of matter and the matter pair’s denominator as the family’s own integer (Section 10.1). *Held to be computable and not yet computed*: the bodies at the closing gap, the moving fixed point of a body and the absorbed fraction with the body’s write; three hypotheses by name are stated where they stand and nowhere else: the vacuum’s write with its one declared number (Section 7.5), the three predictions entered as derived hypotheses (Section 10.2) and the carrier that would mend the lattice’s frame (Section 10.3). One assumption is named with the forms, that a Link is clocked twice by its reading Node’s own clock, $hN = 1$; the clock’s composition and the acts’ factors are derived from the rule, and under the sentence the model’s static metric is the exponential metric, so the bending, the delay and the perihelion are Einstein’s in the clock’s own potential, the sentence itself fixed by those two clicks (Section 3.4); the forms against Kepler’s potential reach Einstein’s only as the matter family’s gap closes (Section 8). One choice was made with a known result in view, and we state it as an input, not a prediction: a click’s share is the square of the meeting, a credit chosen to reach nature’s correlations, so Bell’s $2\sqrt{2}$ follows from it and does not test it. Six numbers are nature’s and not the rule’s: the vacuum content, a declaration; the vacuum write’s rate, Λ , the one declared number of the hypothesis of Section 7.5; the binding range and the proton’s count, tied to each other, the count an input; gravity’s divisor, bounded; and the unit of charge, open (Section 10.2). The magnetic force between moving charges is derived under the holder’s rotation from the rule’s own continuity identity, the sign’s levels one covector, no number matched (Section 8). Table 1 lists every result of the paper with its kind, its fence and its inputs. What is not claimed: that nature is a lattice; that any number of the computed pair [2, 3] is nature’s; that a comparison with nature is more than a consistency statement at the stated order; or that a formula known in the continuum is new where the paper gives its lattice form, the known form standing beside it in every row. Every derivation was checked by two independent hands of the project and, where a world exists, by the implementation’s gates, not by an independent referee; where the text says exact it means exact in the rational arithmetic before the rounding, with the integer remainders bounded as stated; the identifications of Section 10.1 rest on the electron as the free quantum of matter and change with it; and a line decided after a derivation is stated as the law with the implementation’s run pending, a run that differed from its blind being a defect of the implementation to repair and never a result.

How to read the paper. Every claim carries one of six marks. **THEOREM**: proved for every lattice. **DERIVED**: obtained from theorems by argument. **COMPUTED**: a number obtained from the rule’s lines before any run. **ASSUMPTION**: a line held by decision and named where it is used. **HYPOTHESIS**: a line outside the model, under its own name. **EXPERIMENT**: a number the model does not fix, with the experiment named that would fix it; the paper performs none. Every formula carries, beside its mark, its fence. **GAMEBOARD**: a reading of the lattice’s own numbers that no instrument reports, a diagnostic. **CLICKS**: a formula of what a detector reports, the share Q_R/W_c on the forward orbit (Section 5.2) or a ratio of two such, the only kind that nature can be set beside. The algebraic steps of every derivation that the text states without its proof are given in the supplementary material (Online Resource 1), Derivations S.1 to S.32 in the order of the sections, cited as (S.*n*) where used. The runs of the implementation reported in this paper are the gates’ (Section 3.7), the blind written first and the run agreeing with it to the rounding, the engine’s check of the algebra and no experiment against nature. The law’s document, the implementation, the derivations with their checks and the text were worked out with a large language model used as a tool under the author’s direction, and the author reviewed every statement; the Statements and Declarations at the end say how. The paper cites three documents of its archive by the names of their sections: the law [1], the engine [2] and the decisions [3].

The words. The physical location is a *Node*, and its complete local information is its *NodeState*. Nodes are joined by *Links* through six directional *Ports* per Node, and the lattice of Nodes is the *GameBoard* (Fig. 1). A local change is an *Event*, and there is one local rule, Rule3. A *family* is one kind of physical value with its pair and what it holds. Its *record* is its levels over the GameBoard, two per Node with a remainder. A family of quanta has a *dimension*, 1 for one real line and 2 for a plane of two lines. A *body* is the set of Nodes on which a family’s record stands bound, with no law of its own; a *cluster* is bodies standing in one another’s wells and clicking at one another. A *message* is a travelling record laid by the world, such as an emitter’s packet of light. A *detector* is a region of Nodes declared in the file to report, and its *click* is one quantum credited to it from the current it saw through its boundary; the click is the one measurement. An *interval* is one step of the whole GameBoard. A family that holds a value for others to read is a *field family*, or *holder* [1, the objects][2, the words].

Table 1 The results of the paper, each with its kind, its fence and what it rests on. A calibration is a number taken from a measurement of nature; EXPERIMENT is a number the model does not fix, with the experiment named in the text.

Result	Kind	Fence	Rests on	Section
Every run is reversible to the bit	theorem	GameBoard	Rule3	4.1
The gate says MATCH on the two slits', Bell's and GHZ's worlds: $N = 278$, $S = 478/169$, $M = -4$	computed, a run	GameBoard	the implementation	4.1
The rule's acts form no product of levels	theorem	GameBoard	Rule3	4.2
A conserved form, its current, and the momentum flux as the wave's stress	theorem	GameBoard	Rule3	4.3
The count is the record's share; the total share is never negative	theorem	GameBoard	Rule3	4.4
Three exact massive rotations	theorem	GameBoard	Rule3	6.1
Light's speed $1/\sqrt{3}$, its dispersion, and no dispersion along the cube's diagonal	derived	GameBoard	Rule3 for the pair $[1, 1]$	3.3
A click is a meeting, credited by the shares; no collapse; no signal	assumption, derived	clicks	the meeting and the detector	5
A field's reach from its gap, Yukawa's form	derived	GameBoard	Rule3 for a gapped pair	6.4
The stability of a body under the dilation, to first order in the well	theorem	GameBoard	the body's functional	7.2
Free fall is universal for extended bodies of one family	derived, computed	GameBoard	the fixed point and the binding range	7.3
A body's clicks: no light from a steady rotation, light at a breathing body's beat, the resonance at $2\omega_b$, the count's drift	derived	clicks	Rule3 at the body's Nodes	7.4
A cluster: the three levels at a Node as the nesting, three kinds of binding by rank, the click as the only thing between clusters, the recursive hierarchy	assumption, derived	GameBoard, clicks	the universe is bound; Rule3	7.5
The gravitational redshift, $z = f(\omega_0) U$	derived	clicks	the clock's read	8
The bending $4U_b$, the delay, the perihelion's $6\pi U$, $\beta = \gamma = 1$; the shadow's capture radius $2em$	derived under the one assumption	clicks	a Link clocked twice by its Node's clock, $hN = 1$	8
The same forms against Kepler's potential: $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$, $1 + \alpha_K = \tan \omega_0/\omega_0$, closing with the gap	derived	clicks	the fall's formula	8
The moving clock: its fourth order and its limiting speed, Lorentz's as the gap closes	derived	clicks	Rule3's band	8
The force between moving charges, Coulomb's over γ	derived, two hands; the look a check	clicks	the rule's continuity identity, the sign's covector	8
De Broglie, the fall, Newton's pull; the content's index for every family, the lens	derived	GameBoard, clicks	the band and the field's rest	8
Newton's pull, the clock's G and Kepler's G	derived	GameBoard, clicks	Rule3 for $[1, 1]$ and E_g	10.1
$\alpha_{\text{law}} = (3\sqrt{3}/8\pi)/(\Gamma \sin \omega_0)$; the units; the highest photon energy; the gap below 10^{-18}	derived	clicks	the rotation, Poisson's rest, de Broglie, nature's α , LHAASO	10.1
The two-slit fringe row and the lattice's arrival interval	computed	clicks	Rule3 on the declared world	9.2
Bell's $E = \cos 2(a - b)$, $S = 2\sqrt{2}$, $S(\rho)$ and Tsirelson's $2\sqrt{1 + \rho^2}$; every local credit at most 2 (theorem)	fitted; derived	clicks	the credit rule, the meeting's square	9.4
The vacuum content 0; gravity's events at light's speed in every level	assumption; the law's line, built	clicks	declared; every row reads the content	6.5
The binding range, 20 Links	calibration	GameBoard	the proton's 1,836 quanta	7.3
Gravity's divisor, the proton's count, the unit of charge, the matter pair's gap, the Links' wall	experiment	clicks	nature's measurements	6.6, 10.2
The two clocks shift alike; the findings against nature of Section 10.3	derived; against nature	clicks; the clouds' readings	the principle; Rule3	10.3
		GameBoard		

2 Related work

Reversible cellular automata and digital physics. The idea that the physical world is a discrete computation goes back to Zuse [4]; it was developed as reversible cellular automata by Fredkin, Toffoli and Margolus [5][6][7], as a programme by Wolfram [8], and as a deterministic foundation beneath quantum theory by 't Hooft [9][10].

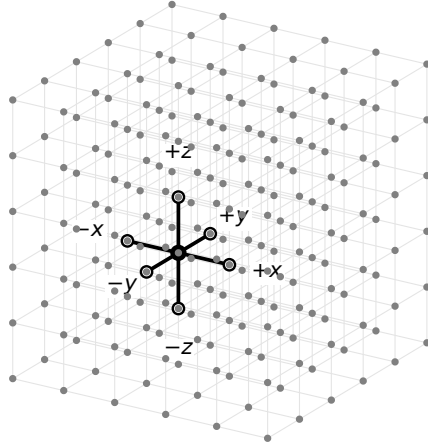


Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it. A Node keeps each family's two levels and its remainders there, and nothing else. Drawn from the definitions; no run.

Quantum cellular automata put a unitary step on a lattice, from the lattice Weyl, Dirac and Maxwell equations [11] and quantum lattice gases [12] to the structure theorems of the reviews [13][14] and the derivation of Weyl's and Dirac's equations from an automaton's symmetries [15]; the present rule's step is an integer recurrence on two levels, not a unitary on amplitudes. Rule3 is a second-order cellular automaton in the sense of Toffoli and Margolus, $a_{\text{next}} = f(a_{\text{now}}) - a_{\text{before}}$ [7], with an integer alphabet and one Euclidean division whose remainder stays at the Node. Feynman noted that a local classical computer reproduces no correlations beyond Bell's bound [16]; the model agrees, and places the excess in the declared credit of the click rather than in the lattice (Section 9.4).

Time-symmetric accounts of measurement. Wheeler and Feynman described radiation as an interaction between the source and the absorber, with advanced and retarded fields together [17]; Cramer's transactional interpretation has an offer wave forward and a confirmation wave backward, meeting at the absorber [18]; the two-state vector formalism evolves one state forward from the preparation and one backward from the outcome [19][20]; Price argued that a time-symmetric account is what Bell's theorem leaves open [21], and Wharton and Argaman showed that every locally mediated reformulation of quantum theory must let some mediating parameters depend on measurement settings in their future [22]. The model belongs to this class, the backward reading from a realised click depending on the detector's setting alone; it differs in that both directions are one integer line, exact to the bit, that their product is a reading and not a wave, and that the choice among the meetings is the one non-local element; the record is an ontic state in the terms of [23].

Bell's theorem and local models. Bell's inequality bounds every local account of a pair's correlations [24], in the form that experiments test [25], and a local model, a joint distribution and the inequalities are one condition [26]. The model's local credits give $S = \sqrt{2}$ and $S = 2$, as they must (Section 9.4); its credit by the contrast reaches the quantum correlation only by discarding pairs, Pearle's road [27], below the efficiency bounds of Eberhard and of Garg and Mermin [28][29], which the loophole-free tests of 2015 close for nature [30][31][32].

Discrete gravity and its tests. Regge calculus [33] and causal sets [34] discretise the geometry of space-time itself; here the lattice is never curved, and the weak-field forms are read from the coefficients (Section 8). In the parametrised post-Newtonian framework [35] the one assumption gives $\gamma = \beta = 1$ exactly in the model's own potential, with the matter band's factors against the potential the tests read closing with its gap. A lattice breaks Lorentz invariance at order one unless its terms are tuned [36], and the bounds are reviewed in [37]; the paper's answer is the closing gap of Section 8.

3 The model

This section states the model: the lattice, the families, the rule, the paces through which one family acts on another, the one write, the readings and the order of acts in an interval [1, the GameBoard and the families; Rule3; the

primitives].

3.1 The GameBoard

The GameBoard is a box of $N_x \times N_y \times N_z$ Nodes. Two Nodes that differ by one step along one axis are joined by a Link, and each Node has six Links, one through each of its six Ports, $\pm x$, $\pm y$ and $\pm z$. Each axis is declared periodic, open or folded. Beyond an open face no Node lies and a level there is read as 0; a folded axis has one layer, and its two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps, which is how a wall with slits is made. A face may also be declared receding (ASSUMPTION): the box then grows by a layer of zeros whenever a nonzero level reaches the Node next to the face, so no event ever meets the face, and a run on such a box is a run on the unbounded lattice. The box grows and nothing stretches, so a receding face gives no cosmological redshift [1, the objects; the unbounded board][3, the receding face].

The group and its action. The symmetry group of the cube has 48 elements, the 24 rotations and, with the reflections, the signed permutations of the three axes, O_h ; with the translations it is the space group of the simple cubic lattice. It acts on the state space by permuting the Nodes and their Ports and carrying every level pair with its Node; the time reversal exchanges a record's present and its past and carries the step to its inverse (Theorem 1); and a rational rotation of a plane record's two parts, $(p, q)/r$ with $p^2 + q^2 = r^2$, carries the rational solutions of the line without its division to solutions and leaves the Wronskian unchanged. Rule3 commutes with the space group's action, since its line reads the six Ports symmetrically and names no axis: a world turned by an element of the group runs as the turned world. Under the group the form D is a scalar, even under the time reversal; the current F a vector, odd under it; the Wronskian W a pseudoscalar; and the click a scalar, since it names a region and an interval and never a Node or an axis. No boost, no rotation beyond the cube's 24 and no scaling is in the group, so Lorentz's group and the continuum's rotations are no symmetries of the lattice, and what the paper finds of the moving clock it finds as a property of the clicks (Section 8) (THEOREM for the covariance and the readings' kinds; GAMEBOARD; S.4) [1, the direction; the four acts][3, the one algebra].

The NodeState holds the model's own numbers and nothing else: for every family and every part, two levels and one remainder, and for every part of a field family, the remainder of its one write. A count, a current, a charge or a tension is a reading computed from these numbers when it is needed; no register, table, formula or random draw is kept at a Node [1, the objects][3, nothing at a Node but the law's numbers].

3.2 Families and records

A family is declared by two keys: its pair [num, den] and its shape. A family of quanta, such as matter or light, declares its dimension, 1 for one real line or 2 for a plane of two lines, and holds nothing. A field family declares its sources, the form, the form with the three tensions, or the Wronskian, which fix its parts, one real line per source, with its divisor E_s and, for the massless field, its rest; a field sourced by the form or the tensions holds the content, which slows the clocks of the families that read it, and a field sourced by the Wronskian holds the sign, the charge's field. A family's record is its levels over the GameBoard, at every Node a level now and a level one interval before, with a remainder; a family of dimension 2 lays two level pairs (re, im), two solutions of the same line, and only a plane carries a charge (Section 5.5). A body is a connected set of Nodes on which a family's record stands bound, and a message is a travelling record laid by the world [1, the objects].

3.3 Rule3

At every Node, in every interval, every family steps by the first line of Eq. (1). The six arrivals through the Ports, each weighted by its axis's coefficient R_a , the Node's own level times S , minus w times the level before, plus the remainder the Node kept, are summed as one whole number and divided once by the wall w ; the quotient is the next level, and the remainder, $0 \leq r' < w$, stays at the Node and never travels. Nothing is rounded away, and the step back is the same division, which is why every run is exact and reversible to the bit (Theorem 1). The name comes from the three axes, one coefficient per direction serving its two Ports, the wall's factor 6 the three directions twice; for light the next level is the sum of the six arrivals over 3, minus the level before [1, the line].

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector $\mathbf{k}$ is $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$. At wave number zero $\cos \omega_0 = \text{num}/\text{den}$, and for the massless pair [1, 1] along one axis $\cos \omega = (2 + \cos k)/3$, so $\omega = |k|/\sqrt{3}$ as $k \rightarrow 0$ (DERIVED, GAMEBOARD; S.1). Light's

speed at long wavelength is therefore $1/\sqrt{3}$ Links per interval, below the causal bound of one Link per interval, and its group speed falls at shorter wavelengths as $1 - 0.083 k^2$ [1, the lattice constants]. In nature's names the vacuum's line is the discrete Klein-Gordon equation exactly: with $S_6 = 6 + \Delta_{\text{lat}}$, $a_{\text{next}} - 2a_{\text{now}} + a_{\text{before}} = c_K^2 \Delta_{\text{lat}} a_{\text{now}} - m^2 a_{\text{now}}$ up to the remainder term $(r - r')/w$, with $c_K^2 = \text{num}/(3 \text{den})$ and $m^2 = 2(1 - \text{num}/\text{den})$, the rest rotation as the mass (c_K is the equation's coefficient; the band's speed c_m of Section 8, (e), is $\omega_0/(3 \tan \omega_0)$, the two one as the gap closes); its slow limit at small wave number is Schrödinger's equation with the inertia $m^* = 3 \tan \omega_0$ and, in a content U , the potential $\omega_0 f(\omega_0) U$ of Section 7.4 (DERIVED, GAMEBOARD); and $m^* c_m^2 = \omega_0$ exactly for every pair: the model's $E = mc^2$ at the family's own light speed, read in three click numbers, nature's its small-gap limit, the departure $1 - \omega_0^2/3$ (DERIVED, CLICKS; S.18, S.21). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, whose dispersion is the known one [38]; what the law adds is one rational $[\text{num}, \text{den}]$ fixing mass and speed together, the equation exact, its departures predictions with bounds (Section 10.2), the integers with the remainder, and the paces (DERIVED).

3.4 The paces: how a field acts on a record

A field acts on a family by changing the family's paces at a Node, the coefficients of Rule3 there:

$$\begin{aligned} p_0 &= \Gamma (1 - 1/\Gamma)^c, & p_i &= \frac{p_0 (c_i)^2}{\Gamma}, & p_a(i, j) &= \frac{p_i q_{ij}}{\Gamma}, & q_{ij} &= \Gamma - t_a(i, j), \\ c &= \sum_{\text{reads}} \text{weight} \times \text{the read family's time level}, \end{aligned} \quad (3)$$

Here c_i is the content the family reads at the Node, the sum over its reads of the weight times the read field's level; the clock p_0 composes, each unit of the level slowing it by the same factor $1 - 1/\Gamma$, to the unit; the Node's pace p_i is the clock twice, p_0^2/Γ , so that the ruler goes with the clock; $t_a(i, j)$ is the Link's tension, one number per Link from its two ends' parts (Section 4.3), read into the Link's factor q_{ij} the same from both ends; and the Link's pace is the product, $\Gamma - 2c_i - t_a$ to first order [1, every row reads the content; the clock is the Node's, the tension is the Link's]. A positive weight makes a hollow, an attraction; a negative weight makes a hill. The pace is the clock's slowing: a record at a Node of pace p rotates and moves as at Γ , with its intervals p/Γ as long. Every family reads every holder of the content at the weight 1, the fields among the readers and a holder of the content its own level among it (the law's line, built; Section 3.7), so $x = (L_b + L_g)/\Gamma$ with L_b the binding field's level and L_g gravity's; a family of dimension 2 also reads the holder of the sign plainly, and light, the holder's own record, reads the holders of the content as every family does and not the holder of the sign; and a family sources what it reads at the weight it reads with, so every pair of families that read each other is reciprocal [1, the paces; the reads are the held row's pair's].

The one assumption of the forms: a Link is clocked twice by its reading Node's own clock. Nothing in Rule3 fixes how a level enters the paces; beside the line the model holds that a level is a count of units. Two parts of the paces follow from that. The clock composes, $p_0 = \Gamma(1 - 1/\Gamma)^c$, each unit slowing the clock as it stands by the same fraction, since the write of one unit is an act and every act is measured in the Node's own units, the read included (DERIVED). And every act of the interval, Rule3's step, the held rows' write and the sign holder's turn, is Rule3's act measured in the cell's units, so a uniform level $U = c/\Gamma$ changes only the units, time by $N = p_0/\Gamma$ and length by $h = p_0/p_a$ per Link, the write carrying $p_x p_y p_z / (p_0 \Gamma^2)$ and the turn p_0/Γ , per proper volume and per proper interval (DERIVED; built, Section 3.7). The one sentence not in Rule3, the model's one assumption of the forms, named as the credit is (Section 5): a Link is clocked twice by its reading Node's own clock, $p_a = p_0^2/\Gamma$, that is $hN = 1$, the ruler going with the clock (ASSUMPTION). The lattice says why twice, a Link having two ends, and not why the reading Node's clock stands at both rather than the two ends' product, which gives $\beta = -1$; the sentence is fixed today by two clicks, the bending's $\gamma = 1$ and the perihelion's $\beta = 1$, and this is where Einstein's $\gamma = \beta = 1$ enter the model by hand. Under it the model's static metric is $ds^2 = -e^{-2U} dt^2 + e^{2U} dx^2$ exactly, the exponential metric [39], Einstein's to the first post-Newtonian order, the shadow's capture radius $2e m$ its second-order click against Schwarzschild's $3\sqrt{3} m$ (Section 8), in the lattice's frame, the frame a second residue not derived from Rule3 and read by the binary pulsars' α_1 (Section 10.3). The quadratic $(\Gamma - c)^2 + c^2$ is the composition's second-order truncation [1, the same clock and the same ruler for every act; the paces compose][1, what is open, item 15].

The band at a pace. From the coefficients, the rotation of a record at a Node of clock pace p_0 and Link paces p_a is (DERIVED, GAMEBOARD; S.1)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (4)$$

The mass term scales with the clock's square and the band term with the Link's. Two readings follow. First, at $k = 0$ every massive band's rest rotation obeys $1 - \cos \omega_b = (p_0/\Gamma)^2(1 - \cos \omega_0)$, the clocks' shifts of Section 7.4. Second, along a Link every wave's speed falls with the Link's pace, $\Gamma - 2c - t_a$ to first order, so light is delayed and bent by twice the clock's share. At a Node of content $x = L/\Gamma$ light's band is $[2 - 4N^4, 2]$ in $2\cos \omega$ with $N = (1 - 1/\Gamma)^L$, and at long wavelength $\omega = N^2 |k|/\sqrt{3}$: light's speed is $N^2 = e^{-2x}$ times its vacuum speed, to the unit, an index $n = e^{2x}$, $1/(1 - 2x)$ to the first order, with $n - 1 = 2x$ (S.2). A record whose rotation lies outside the band at a Node is evanescent there and reflected whole; light meets a total mirror where $p_a < \Gamma \sin(k/2)$ [1, the band at a pace; the rows against nature, (b)].

The guard. A pace is positive by the composition and must lie below an edge P , with $P^2 = 2 \text{den} \Gamma^2 \text{div} (\text{den} + |\text{num}|)$, which is Γ^2 for light: beyond P the mode at wave number π on every axis grows without bound, and at or below P it stays bounded (THEOREM, GAMEBOARD). The guard is the Courant, Friedrichs and Lewy condition of the explicit scheme Rule3 is at the vacuum's pace [38], checked once at load. Under the composed paces no Link ever closes: the clock $\Gamma(1 - 1/\Gamma)^c$ is positive at every content and the Node's pace is its square over Γ , so a deep well freezes a clock and closes nothing, and the quadratic form's horizon at $\Gamma/2$ is a feature of the truncation and not of the law (DERIVED, GAMEBOARD; S.3). A Node whose clock is frozen to the unit is cut from its six neighbours: no current crosses its Links and its share is not a number, so it is dark. The guard's bound, $|2\cos \omega| \leq 2$ at the band's edges, admits the edge itself, where a linearly growing solution exists, and it is stated for $\text{num} \geq 0$; at that edge the vacuum's checkerboard mode, driven by the remainders, grows as $t^{3/2}$, no click reading it, and reaches the loader's bound near 1.4×10^5 intervals at $\Gamma = 6,000$ (COMPUTED, GAMEBOARD), a negative numerator's lowest rotation lying at $k = 0$ (S.3). At a negative content light's zone-corner mode grows, so light refuses a content below 0, and the vacuum's content is 0 (Section 6.5) [1, the guard; the frozen Node has no share].

3.5 The write, the readings and the interval

The one write. Every write into a field's level is one act, the second line of Eq. (1): at a Node in each interval a held part gains $(\sum_f w_f q_f + r) \text{div } E$, one division per part with one remainder r carried at that Node. Here q_f is the sourcing family's own quantity at the Node: its form D_f for a holder of the content, its Wronskian W_f for the holder of the sign, or its tension $T_{aa,f}$ for an axis part. w_f is the weight with which that family reads the written field, and E is the part's wall, $E_s T$ for the time part and $E_s W_c$ for an axis part. The sources are summed in the numerator and never divided apart, so no remainder stands per sourcing family [1, the primitives][3, the one algebra].

The readings. A reading, which we also call a booking, is a bilinear form of the levels with declared coefficients; it writes nothing on the GameBoard. The law reads three, the third line of Eq. (1). The form $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$, the present against the product of the future and the past, is $B^2 \sin^2 \omega_b$ at every interval for a standing record of peak B ; its quotient over T , the well, sources the fields. The Wronskian W_i is the charge (Section 5.5). The current $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ into the Node i from its neighbour j is antisymmetric and, with the Link's factor q_{ij}^2/Γ^2 , is the Link term of the conserved form (Theorem 3), the plain current where no tension stands; a wave travelling from i to j has $F_{ij} < 0$. A static record has $\text{now} = \text{next} = \text{before}$, so its form and its current vanish: a static field carries no quanta and no detector sees it [1, the booking; the conserved form].

The interval. Every read is of the values at the interval's start; Rule3's line reads one Link per interval, and nothing in zero time; every booking of the interval, the step, the tensions and the hold's write, is read from the record as it stands at the interval's start, so a Node's NodeState after one interval depends on the Node and its six neighbours alone, the interval's dependency radius one Link (the law's line, GAMEBOARD) [1, the interval's dependency radius is one Link]. The interval has four acts in order: (i) the read, each family's paces from the field levels, the reading field's own among them, and each Link's factor from its tension; (ii) Rule3 on every record; (iii) the readings, each record's currents, tension, form and Wronskian at the levels the step started from, and each detector's report; (iv) the write of each field from its summed sources. [1, the interval][2, the interval].

3.6 The postulates

The model rests on seven postulates [1, the postulates]: (1) the world is Nodes and Events, an Event a level stepped by Rule3 with its remainder, a click a detector's report, and nothing else happens; (2) a Node holds bounded local information, read from itself and its six neighbours, and nothing of the past is kept; (3) consistency is local and causal, one Link per interval for every booking of the interval (Section 4.3), with one exception, the credit of a click by the shares over every declared detector, the one non-local element, not derived; (4) the causal speed is one Link

per interval; (5) every physical calculation is on bounded integers, with no floating-point number, no root and no random draw in the law; (6) a measurement is a detector's click, the net current into a declared region through its front boundary over its window, never a Node; (7) the GameBoard, where no one measures, is deterministic and reversible to the bit, and the detectors' clicks are the only thing claimed to represent nature. The line and its clock are fixed: every mend of the model is a family, a holder, a declaration or a read, never a change of Rule3 (ASSUMPTION) [3, nothing that is not Rule3 remains].

3.7 The implementation

The implementation, Universe24 [40][2], is the law's implementation and nothing else: a GameBoard of Nodes, every family's NodeState at every Node, and one interval of acts in the law's order, each a call of Rule3 on whole-board arrays of integers at a declared width, every neighbour read through a Port. It holds no number of physics, no formula, no family's name and no flag, every physical value coming from the run's files; a run is headless, writes the click lines, the measurement, and runs for results only after the gate of Section 4.1 says MATCH [2, the engine; the NodeState; the output; the gates]. It holds the law's lines of Section 3.4, the composed paces, the Link's factor, every row at its reads' paces, the write's and the turn's factors and the guard's two sides; the gate worlds' outputs under them are bit-identical to the quadratic forms' runs [2, the NodeState; the interval]. One blind of the law awaits its three-dimensional run, the one-dimensional chain not carrying the ruler: a body in a uniform level, a well of at least 300 levels at $U = 0.05$, its rest line's and its beat's $2 \sin(\omega/2)$ ratio to N within 1.5 over the well's depth in levels, an absolute bound [1, what is open, item 29]. The gates' runs, the blind written first and the run agreeing with it to the rounding: the two slits, $N = 278$ against the law's real line's 277.8 (278.0 in the physical labels), the Huygens blind 273 written first, the real line computed by the script `paper/general_formula/two_slits_real_line.py`; Bell, $S = 478/169$ with the marginals $1/2$; GHZ, the eight shares, every pairwise E 0 and $M = -4$; and the back-in-time gate MATCH on every world (COMPUTED, runs of the implementation; CLICKS) [1, the rows against nature, (g) and (h); the GHZ gate]. The paper reports the gates' runs and derives the rest: a run of a declared world computes what the algebra of that world states, the implementation held to the law's lines by its tests and the gate.

4 Exact properties

This section proves what holds exactly on every lattice: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current and its momentum flux, and the count as a reading of the record.

4.1 Reversibility

Let σ be the direction, +1 forward and -1 backward; let div and mod be the floor division and its remainder in $[0, w)$; and let ρ be the remainder carried in. With

$$u = \sigma \left(\sum_a R_a \text{arr}_a + Sx - wy \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (5)$$

the forward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and the backward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Theorem 1 (the direction). *The backward call returns (a_{before}, r) bit for bit; the reversibility of every step needs no second function.*

Proof. Forward, $\sigma = 1$, the line is the definition of div and mod . Backward, $\sigma = -1$: write Y for the right side's first two terms, $\sum_a R_a \text{arr}_a + Sa_{\text{now}}$. Then $u = w a_{\text{next}} - Y + r'$ and $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$. From the forward line, $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$ with $0 \leq r < w$, whose ceiling divided by w is a_{before} exactly; and $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$. [1, the direction]. $\square$

The whole run therefore goes back: N intervals forward and N back return every array of every family bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds (THEOREM, by construction from Theorem 1). The implementation checks this as a gate, running a world forward and back and comparing every array at every interval; on the worlds of Section 9 it reports MATCH over their whole runs (COMPUTED, GAMEBOARD) [1, the inverse; the objects][2, how to run a world][3, the Node knows no family]. The condition rules out one read: a coefficient read from a Node's own state across a step cannot be recovered at the

step back, so a record's rotation sense is never read at a Node and the holder of the sign is read by the record's shape, a plane or a line (Section 5.5) [1, the interval][3, the board goes back in time, as a gate; there is no sense].

4.2 What the acts can form

Rule3's line is one act, and declared coefficients make it four: (a) the read, the line with its six arrivals; (b) the one-Node step, the line with the arrivals declared 0, which rotates a level by Chebyshev's recurrence; (c) the division, the line with $R = S = 0$, Euclid's division with the remainder kept; and (d) the load, a declared integer in the numerator, which is how a count enters a level [1, the four acts].

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

Proof. Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of such functions are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product $\text{now}_i \text{before}_j$ in the pair $(\text{now}_i, \text{before}_j)$ is 1. So nothing bilinear comes from the acts alone. $\square$

So every product of two levels is a reading and never a write, and the three readings of Section 3.5 are the only products the law takes. A line enters the law only if it passes three tests: it is generic (one primitive with declared integers, no family's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers). A line that needs more is a hypothesis under its own name. No act of the interval takes a square root; the integer square root appears only when a world is laid, as the fixed point of the division act $x \leftarrow (x + n \text{ div } x) \text{ div } 2$ [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

4.3 The conserved form and the momentum flux

The read coefficient $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$ carries the Node's factor and the Link's; the Node's factor is not symmetric where the paces differ and the Link's is, so the conserved form carries the weight $1/p_i^2$ on its Node terms and the Link's factor q_{ij}^2 / Γ^2 on its Link terms:

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{ num } \sum_i \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} \text{now}_i \text{before}_j. \quad (6)$$

Theorem 3 (the conserved form). *At constant paces for which a positive Node weight d_i with $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$ on every Link exists, E with the weights d_i is exact under the line without the rounding. Under the law's read coefficient, the Node's pace squared times the Link's factor squared, $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$ with $q_{ij} = q_{ji}$ and $p_i = p_0(c_i)^2 / \Gamma$ the composed Node factor, the weight $d_i = 1/p_i^2$ cancels the Node's factor on both sides of every Link, so E is exact for any field and any tensions. With the rounding each Node's remainder adds at most one unit of the level per interval to the walk of E . Its Link term carries the Link's factor, so the current of Rule3's own conservation law through the Link ij is $\text{num } (q_{ij}^2 / \Gamma^2) (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$, the plain current F_{ij} where no tension stands, $q_{ij} = \Gamma$; the form with the plain current holds there alone, and with a tension it is not conserved (two witnesses, S.32).*

Proof. Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ with $\mathbf{M}$ the matrix whose diagonal is S_i/w and whose entries on the six Links are R/w . Let $\mathbf{D}$ be the diagonal matrix of the weights $1/p_i^2$; then $\mathbf{DM}$ is symmetric iff $d_i M_{ij} = d_j M_{ji}$ on every Link, $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$: with $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$ the weight $1/p_i^2$ cancels the Node's factor on both sides of every Link and the Link's factor q_{ij}^2 is the same from both ends, so $\mathbf{DM}$ is symmetric for any field and any tensions. For a symmetric $\mathbf{K} = \mathbf{DM}$ the quantity $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$ satisfies $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$: with $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$, $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$, which is the other side. Multiplied by w this is E up to the Link term's sign convention [1, the conserved form]. $\square$

E is a diagnostic. Where the paces move, the total form $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$ changes exactly by $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$, which is 0 at fixed uniform paces and negative where a well deepens under a standing record (THEOREM, by substitution; GAMEBOARD) [1, the conserved form].

The momentum flux is the wave's own stress. The momentum density along the axis a at the Node i is the current per axis, $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$, and the line without its division gives exactly, at constant paces, the identity below; the integer step adds the remainder term $\delta_i \Delta_a(\text{now})_i - \text{now}_i \Delta_a(\delta)_i$ with $\delta = r - r'$ and Δ_a the centred difference, which the implementation's test includes (THEOREM for the line, GAMEBOARD),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i-b) - G_{ab}(i)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

with $G_{ab}(i)$ the stress reading on the b -Link $(i, i+b)$, the levels now alone. The proof substitutes the line into $w P_a(t+1)$: the S term cancels, the w term is $w P_a(t)$ and the R term is the difference of the fluxes. The tension on the axis a at a Node, the part a field reads into the Link's pace and the source of a field's axis part, is the momentum flux averaged over the Node's two a -Links,

$$T_{aa}(i) = -\text{num} (G_{aa}(i) + G_{aa}(i-a)) \text{div } 2, \quad (8)$$

$$G_{aa}(i) = \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}),$$

and each Node sources its own part, $-\text{num} h_a(i)$ with $h_a(i) = \text{now}_{i-a} \text{now}_{i+a} - \text{now}_i^2$, read from its two neighbours along the axis at the interval's start; the Link's tension is its two ends' parts, one number per Link read the same from both ends by the pace at the next interval. A plane wave of amplitude b at the wave number k has $G_{ab} = -2b^2 \sin k_a \sin k_b$, so the flux along the wave is positive, as a wave's pressure is (S.7); a packet of count c and velocity $\mathbf{v}$ has $c v_a v_b$, the stress of dust. With the sign as written a wave's stress deepens the content around it, so light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the record's own stress].

4.4 The count is the record's share

A family's count of quanta is a reading of its record and no line of its own. At every Node the record's share is

$$e_i = 3 [w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i] \text{div} \left(2 \sum_a p_a^2 \right) - \text{num now}_i S_6(\text{before})_i, \quad (9)$$

the conserved form's Node term over the mean of the three Link paces squared, less the plain Link term, read in quanta over W_c as the nearest whole number. Read with the vacuum's coefficients instead of the Node's own paces, the same expression is the vacuum's share, the unit of the tables of Section 7. Over one step of Rule3 the share's rational value, the weighted Node term before its floor, changes by exactly the six currents into the Node, plus a term that vanishes where the three Link paces are equal, plus Rule3's own rounding (THEOREM for the rational identity, GAMEBOARD); the integer share of Eq. (9) adds the reader's own floor, a residual strictly between -1 and 1 per Node and level pair, distinct from Rule3's remainder:

$$e_i(\text{next}, \text{now}) - e_i(\text{now}, \text{before}) = \sum_j F_{ij} + \text{num} (\text{next}_i - \text{before}_i) \sum_a \frac{(3p_a^2 - P^2) \text{arr}_a(\text{now})_i}{P^2} - \frac{3(\text{next}_i - \text{before}_i)(r'_i - r_i)}{2P^2}, \quad (10)$$

with $P^2 = \sum_a p_a^2$, arr_a the two arrivals of the axis a summed, and r, r' Rule3's remainder before and after the step. So over any region whose three Link paces are equal, the total share changes only by the currents through the region's boundary, the remainder terms and the readers' floors (THEOREM, GAMEBOARD) [1, the count is the record's share; the share's change is the currents]. A formulation moving whole quanta by a division of the current would put numbers at every Node that no act of Rule3 reads back; the law does not carry it (ASSUMPTION) [3, the count's line and the sense's line leave the law]. Where the paces move the share drifts with no current crossing any Link, Eq. (13), a diagnostic [1, the paces' own change moves the share with no current].

The signed integers. The levels are signed integers, the currents are signed by their direction and the Wronskian by its sense, and a click is one whole quantum, never negative. The total share of a closed or periodic GameBoard is non-negative for $|\text{num}| \leq \text{den}$, with equality for a static uniform level of light and, on an even periodic board, for its checkerboard (THEOREM, GAMEBOARD; S.6). One Node's share can be negative at a lay's edge, and a region's by at most half a level-square per boundary Port. Over the whole passage of one forward packet the inflow accumulated

at a region is non-negative up to the rounding, since the law's current is forward at every Link, so the credit by the shares is well defined; where a window is cut inside a passage the instrument credits $\max(s_R, 0)$ (DERIVED, CLICKS) [3, the law's integers are signed].

5 Measurement: the click as a meeting

The click is the meeting of the future with the past at a declared detector, and no act of the GameBoard. This section states what is on the lattice, the meeting, the detector, what a click leaves behind, the charge and light, and what locality costs [1, the click; the reading].

5.1 Events, and no waves

On the GameBoard there are events and amplitudes. An event is a level split by Rule3 to the six neighbours; an amplitude is the level Rule3 produces at a Node, the sum of the split events there; and the record is the amplitudes over the GameBoard. In our world there are clicks, the detectors' reports, and nothing else. In the two slits the emitter's events pass both gaps, every path of the future at once; the fringes are where split events from the two gaps meet at the screen with the same sign. The past is made of the same split events: read backward from the realised click by the same line, Rule3 at -1 , they form the past's tree, which also passes both gaps back to the emitter. What stays on the GameBoard, the amplitudes no click realised, is not in our world (ASSUMPTION) [1, no waves on the event board][3, no waves on the event board].

5.2 The click is the meeting of the future with the past

Assumption 1 (the click is a meeting, the law and not a hypothesis). A record runs on the GameBoard in both directions by the one line: forward from the emitter, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at -1 . A click is where the two carry the same signal at a declared detector; its share is the product of the two directions at the region, the form $D = \text{now}^2 - \text{next} \times \text{before}$, the now against the future times the past; which meeting is realised is the draw, the one non-local place. The detector only asks whether the future and the past met at it.

Computationally, by the linearity of the line, the backward reading from a realised click equals a sum over the forward records, so the meeting adds a reading of what a click is and no mechanism of choice; which meeting is realised stays the draw by the shares, the one non-local place.

In plain words (Fig. 2): a click happens when the past meets the future; the click is the now; the detector only asks whether the future and the past met at it; and a click is created only at their meeting. This is a line of the model and not a hypothesis [3, the click is the meeting of the future with the past].

The click, algebraically. Let X be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step S_t is Rule3 over the interval with the paces read from the fields at its start; it is a bijection of X , exact by the division with its remainder; the line without its division is linear in the levels at fixed paces, the linearity the backward reading uses, while the integer map is not affine; its forward orbit is the future and its inverse orbit the past. The readings D , F and W are bilinear forms on X , covariant under the symmetry group of Section 3.1 and never written. An instrument R is a declared region with its front Ports and its coefficients, which define Q_R : for one line a quadratic form of the levels, the inflow through its front Ports accumulated over the window; for a pair, the window's sum of the squared cross-side sum over the parts, quadratic in the joint amplitude, the product of the two sides' levels, and so quartic in the levels. A click is the pair (R, t) , the credit of one whole quantum to the instrument R at the interval t , drawn with the probability $s_R / \sum_{R'} s_{R'}$, where $s_R = Q_R(x_t) / W_c$ is the instrument's form on the forward orbit over the quantum's measure of a current. In the meeting's words, the inverse orbit from the credited state coincides with the forward orbit on the realised path, and the share is the bilinear form's value at the coincidence. The click writes nothing to X : it is the one map in the model that is not a bijection, from the vector of shares to a unit vector, and it is a scalar under the group, so no click names a Node. Every observable is Q_R / W_c on the forward orbit or a ratio of two such. The window is the whole passage of a packet, or the credit floors at 0 (Section 4.4); a Node whose three Link paces are 0 stands in no instrument's front and is never clicked (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS) [3, the click's algebraic definition].

One form for every click. At the credit the parts of a record are read with the detector's coefficients, each part paired with the same part through the common origin, summed and squared; the share is that density, and the

draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's gate is the case through the common origin: a pair born as one event is two trees with one root, and the backward reading from the first click passes the root and continues forward on the partner's tree (Section 9.3). In the implementation the joint share is computed from the forward records alone, by the linearity of the rule; no backward Rule3 is stepped, and the gate of Section 4.1 shows that the backward reading exists to the bit (DERIVED, CLICKS).

No duality. What is called the particle as a wave is the future: the record run forward by Rule3 with every possibility, which has not yet met the past. What is called the particle as a particle is the present: the click, one whole quantum credited at one region in one interval. The particle's track is the past: the chain of realised clicks, and the same record read backward through them. There are not two things but one record read in two directions, and the point where the two readings coincide is what appears as a particle. After a click no wave vanishes and nothing collapses: the record goes on, the click is a credit, and the past has grown by one click (ASSUMPTION) [3, the wave is the future, the click is the present, the track is the past].

5.3 The detector is a declared instrument

Assumption 2 (the detector's declaration). A detector is one connected region of Nodes declared in the file, never one Node, at least half the wavelength of the family it reports across the beam (q/p Links for the wave $[p, q]$, 4 at $\lambda = 8$); a narrower region, a single Node or a region in pieces is refused at load. It is opaque: backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns, with no act at its Nodes and no log. Its click is its report, per interval and family: the net current into the region through its front boundary Ports, the Ports leading in from the declared board, summed in integers, with the region's name and the family and never a Node; one wall $W_c = 3 \text{ den } T$ of the inflow over a passage is one quantum, so a screen saw N quanta where its regions' inflows sum to $N W_c$. The credit: one quantum to one detector, drawn by the shares of the regions' inflows, read outside the GameBoard over every declared detector of the world, one credit per quantum and never two. The credit is the one non-local place of the description and no line of the law.

Why a region: a record confined to one Node contains every wave number at once, $\Delta x \Delta k \geq 1/2$ on the lattice as in nature; so no click names a position finer than a region, and every per-Node number is a diagnostic [1, the click is the detector's report; no click names a Node][3, no click names a Node].

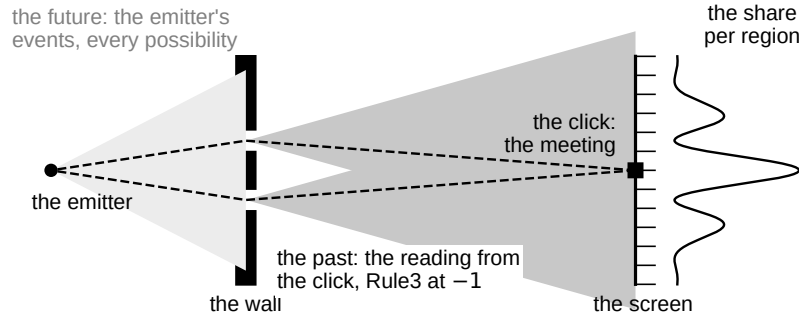
What produces a click is deterministic: the meeting, the current of the amplitudes through a region's boundary. What is not known is which of the possibilities becomes the click: when one possibility is spread over several detectors, the GameBoard gives the shares, no line of the law gives which detector fires, and the instrument draws. Exclusivity, one click per quantum, and Born's rule, the credit's chance equal to the intensity's share [41], are therefore declarations of the instrument and not findings of the model [1, what produces a click is not known]. The uncertainty relation is not a separate rule: it is the detector's three declarations, a region of at least half a wavelength, a credit per interval and one draw per quantum (ASSUMPTION) [3, the uncertainty relation is not a rule of its own]. What each declaration costs: a region of n Nodes reads a fringe of period Λ with the factor $\sin(\pi n/\Lambda)/(n \sin(\pi/\Lambda))$; a frequency is read to $\Delta\omega \geq 2\pi/N$ over N intervals; and N quanta drawn by the shares scatter as $\sqrt{N p_i(1 - p_i)}$, so a pattern is in the histogram and never in one click (DERIVED, CLICKS; S.8) [2, the output].

5.4 The click ends nothing

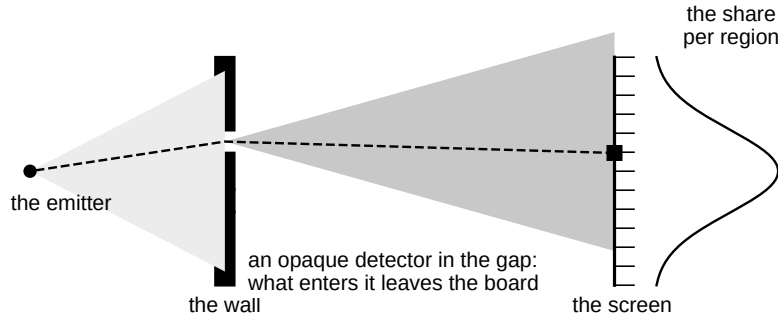
Assumption 3 (the click ends nothing). The click is a detector's reading, local to its boundary; the record goes on after it and nothing ends anywhere. What a detector absorbed leaves the declared board through the face that backs it; the other possibilities go on splitting and leave through the receding faces, events that were not realised, in no click. The ending of a record at a click by the detector is decided against and is no hypothesis of the law.

This is the model's claim and not a deficiency: perhaps nature is local. No signal passes, since nothing at one detector changes the amplitudes at another. A measurement needs no collapse: the record never ends, the click is a reading, and the backward reading from the click writes nothing [3, the click ends nothing]. A bound body can act as an instrument in the same form: the future is light's forward record at its boundary, the past is the body raised by one quantum read backward to the source, and the meeting is the same frequency, the resonance of the body's rotation with the light's (DERIVED, CLICKS; Section 7.4). The body's write back waits on the fixed point with the write [3, a Node of the board is read only through a detector][1, the rows against nature, (y)].

(a) two gaps: the share alternates



(b) one gap closed: one gap's envelope



(c) a pair: the meeting through the root

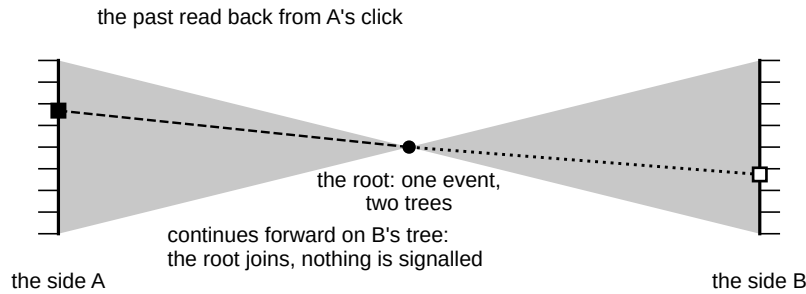


Fig. 2 The click as the meeting of the future with the past, schematically (ASSUMPTION; drawn from the definitions, no run). (a) The two slits: the emitter's events spread forward through both gaps, every possibility at once (the grey fans, the future); from a realised click at a region of the screen the same line is read backward, Rule3 at -1 , through both gaps to the emitter (the dashed lines, the past); the share along the screen, the product of the two directions, alternates. (b) One gap closed by an opaque detector: its events leave the declared board; the backward reading meets one gap and the share is one gap's envelope. (c) A pair, two trees with one root: the backward reading from A's click passes the root and continues forward on B's tree (the dotted line); the root joins and nothing is signalled.

5.5 The sign is the rotation sense

A family of quanta laid as a plane carries its record as two level pairs, (re, im) , stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a field's parts stay real. The charge density at a Node is the reading

$$W_i = re_{now} im_{before} - im_{now} re_{before}, \quad (11)$$

the Wronskian of the two pairs, which the line without its division conserves exactly with the conserved form's weights, under Theorem 3's condition on the paces (THEOREM, since **DM** is then symmetric; GAMEBOARD; S.9).

The integer step adds to each part the remainder term $\epsilon = (r - r')/w$, $|\epsilon| < 1$, so the integer W changes at a Node by $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$ per interval, not zero in general: at the vacuum's pace of [2, 3] the parts (1, 3) with zeros before step to (1, 4) and W from 0 to -1 (COMPUTED, GAMEBOARD; S.9); an exactly conserved integer charge is open (Section 10.3). Its local change is the Link current num $(q_{ij}^2/\Gamma^2) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$, and its total over the GameBoard is invariant under any change of the paces that keeps them uniform at each Node, since the bracket $\text{im}_{\text{now}}^\top \mathbf{M} \text{re}_{\text{now}} - \text{re}_{\text{now}}^\top \mathbf{M} \text{im}_{\text{now}}$ vanishes for every symmetric $\mathbf{M}$ (THEOREM, GAMEBOARD). A real record ($\text{im} = 0$) has $W = 0$, so light, the real record of the holder of the sign, is neutral; a rotating record has W of either sign by its sense [1, the sign is the rotation sense; the sign's count is the record's own].

Who reads the holder of the sign is decided by the family's dimension and by no name (ASSUMPTION, GAMEBOARD): a family of dimension 2 has a Wronskian and reads it, a family of dimension 1 has $W = 0$ and does not. The holder of the sign does not read its own level, so light does not read it and a static charge does not bend light. A reader's own sense never enters its coefficients, since a coefficient read from a Node's own state would break the exact inverse (Section 4.1); so the holder of the sign acts on its readers by the act its row declares: **rotation**, the holder's level over Γ as the angle θ turning the two-part record in its own plane, $e^{-i\theta} z_{\text{next}} + e^{i\theta} z_{\text{before}} = (S z_{\text{now}} + \sum_a R_a (z_{+a} + z_{-a}))/w$, three exact integer shears inverted bit for bit with θ read at now alone, under which like and unlike senses are told apart by the record's own sense; which sense moves apart is one bit the model does not fix, declared by nature's click that like charges repel, the shears' exact angle $2 \arctan(L/2\Gamma)$, L/Γ at small angles, and the magnetic force follows as the Link's phase, Peierls' coupling (Section 8) (ASSUMPTION for the declared bit, GAMEBOARD) [3, there is no sense; every family has a dimension; the rotation round, built][1, the sign holder rotates the two-part record; the rows against nature, (f)].

5.6 Light is born by the write

Light is the holder of the sign's own record: one record whose time part is a real level pair, stepped by Rule3 at the paces of its reads, so that light is bent by the content, and written by the one write from the Wronskian of the records that read it. A static rotating body writes a static level, Coulomb's field, and radiates nothing. A breathing or moving body writes a varying source at its Nodes, the field carries it away at one Link per interval, and those spreading events are the light (DERIVED, CLICKS). A detector reads light's current through its boundary in quanta of W_c , and the travelling events of the other fields, which in gravity's case nature calls a gravitational wave, enter a detector the same way. A body's quanta that leave it stay its family, and no quantum changes family at a crossing. A body reads the holder of the sign at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write].

5.7 The price of locality

Two consequences of the model's locality, stated against nature's clicks [1, what is open, item 4; the rows against nature, (g) and (h)][3, the click is the meeting of the future with the past].

- (i) *Exclusivity and Born's rule are declared.* A quantum spread over several detectors is a set of shares of the current at each; the model gives the shares and the instrument gives the click.
- (ii) *Bell.* A pair is one laid record, a common cause, its phase a shared classical variable inside Bell's hidden variable [24]; every local credit of that phase gives $|S| \leq 2$ (THEOREM, CLICKS; Section 9.4) [25], and the meeting through the pair's root gives $E = \cos 2(a - b)$ and $S = 2\sqrt{2}$ with nothing signalled, resting on the credit rule, a declaration (ASSUMPTION, FITTED; CLICKS). [3, the click is the meeting of the future with the past].

A which-way measurement without absorption is no third price: nature has no detector that marks a path without a click, the model's opaque detector in one gap loses the fringes as nature does, and a bare region's inflow is a GameBoard diagnostic and never a measurement [3, go over the paper and see which mismatches with nature the clicks resolve].

5.8 The formulas of clicks

Every observable of the model is Q_R/W_c on the forward orbit or a ratio of two such (Section 5.2). Table 2 collects the formulas of clicks the model gives, each with its status and the section that derives it [3, the click's algebraic definition; the mathematics of the bound bodies and how it bears on the clicks].

Table 2 The formulas of clicks, each with its status. Every row is a formula of what a detector reports, CLICKS. U is the clock's share and U_K the Newtonian potential the orbits measure (Section 8).

The formula	Status	What the model gives	Section
Born's rule, the count at a region	derived	$N_R = Q_R/W_c$, the share of the inflow through the region's front over the window; the pattern is in the histogram, $Np_i \pm \sqrt{Np_i(1-p_i)}$	5.2, 5.3
The dead time	declaration	not derivable: the shares credit a region any whole number of quanta in one interval, and W_c over the inflow's rate is the mean spacing of the clicks and not a least one	5.3
The two-slit row	derived; computed	the near-field minima at the rows 15 and 33; the screen's inflow peaks at the interval 68.0 against the fourfold refined lattice's 61.8	9.2
The arrival and its spread	derived; computed	$t = L/v_g$ with $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega$; the click's duration $\sigma_t = \sqrt{\sigma_x^2 + (\omega'' L \sigma_k / v_g)^2} / v_g$ from the band's curvature ω'' , 4.13 intervals at $k = \pi/4$ over $L = 34$ Links, with $\sigma_x = 1.70$ Links, $\sigma_k = 0.302$ per Link, $v_g = 0.547$ and $\omega'' = -0.0794$	9.2
Bell's pairs	the credit declared (Born's rule for a pair); E and S derived under it	$E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ and $S = \sqrt{2}(1 + \rho)$, 478/169 at the integer settings for equal parts	9.4
GHZ, three quanta	the same declared credit; E_3 and M derived under it, read by the engine's gate exactly	the same credit on four lines: $E_3 = \cos 2(a + b + c)$, every pairwise E 0, Mermin's $M = -4$ where every local credit gives $ M \leq 2$	9.4
The clock's click rate	derived	$1 - f(\omega_0)U$, $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, $1 - U$ at a small gap, every splitting and the beat alike; a cavity of bodies the same, declared faces N^2	7.4, 8
The bending's clicks	derived under the one assumption; in U_K	$4U_b$; $(1 + \text{den}/\text{num}) 2GM_K/(c^2 b)$ against Kepler's potential	8
The absorption line	derived (the read's side); computed when the fixed point with the write is	parametric at $2\omega_b$, the growth $\epsilon/(4 \sin \omega_b)$ and the width $\epsilon/\sin \omega_b$; the pair line at $\omega_i + \omega_j$ of which it is the degenerate case; the transfer line at $\omega_j - \omega_i$, Rabi's rate, two derivations	7.4
Emission	derived	a steady rotation and a uniform motion write no light; a breathing body writes at its beat, at the first order in the breathing's depth	7.4
The which-way sum	derived	the continuity of the one inflow through the two fronts: a region behind one gap reads that gap's share and nothing of the fringe	5.7
The uncertainty relation	declaration, with its inequalities	$\Delta y k \geq \pi$ from the region, $\Delta \omega \geq 2\pi/N$ from the interval, $\sqrt{Np(1-p)}$ from the draw	5.3
No signalling	derived	$P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$, independent of b for parts of any amplitudes	9.4
A body's count in clicks	derived; the click's whole declared	the share summed over the body's Nodes, what a full absorber credits; under a moving well it drifts by Eq. (13); a body's click one transition, $\hbar\omega$ per click	7.4
Magnetism	not derived under the plain read; derived under the holder's rotation, two hands; the engine's look a check of its booking	F_C/γ : the sign's halved current over the wall den T , no number matched; 0.995 at 0.1c and 0.866 at 0.5c; under the plain read $(1 + 1.5\beta^2) F_C$, the wrong sign	8

6 The families and the universe

A family is a band of the one rule and not a separate law. Its row in the universe file holds only what the rule and the geometry do not fix: its pair [num, den], with $\cos \omega_0 = \text{num}/\text{den}$, and its shape: for a family of quanta its dimension, for a field its sources with its divisor. A pair written over the GameBoard's one denominator Γ makes a mass one integer m , with $\cos \omega_0 = m/\Gamma$ and the wall $6\Gamma^3$; a reduction to lowest terms is a reading and never an act of the loader, so such families share one resolution of their remainders [1, the families from the rule; the mass is one integer; the wall is one].

How a family enters. A family enters the world as one row of the universe file with two keys and nothing else: its pair, and its shape, which for a family of quanta is its dimension, 1 or 2, and for a field its sources, the form,

Table 3 The rule’s own universe, its four families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and its shape): each family with its pair [num, den] (in brackets over $\Gamma = 6,000$), what it holds at its divisor, its shape, what it reads, and its free number. A sixth family, the pair family of light, two real lines never summed, is declared by Bell’s world and not by this file (Section 9).

Family	Pair	Holds, at the divisor	Shape	Reads	Its free number
gravity, the time part and the three tensions	[1, 1] ([6000, 6000])	the content, E_g (at least 10^6 at $\Gamma = 6,000$, a click bound from the two slits’ screen; 1,000 in the bodies’ worlds) the sign, 1	1 + 3 real lines, sourced by the form and the three tensions	the content, its own level among it (the law’s line, built)	E_g , the hierarchy; its rest c_{vac} , 0 by declaration
the charge, the holder of the sign, whose quanta are light, born by the write	[1, 1]		1 real line, sourced by the Wronskian of its readers	the holders of the content, at the weight 1; never its own level	none
the binding field, not the strong force (Section 10.3); its own events have the gap	[2400, 2401]	the content, 1	1 real line, sourced by the form	the content, its own level among it (the law’s line, built)	the pair, its range $R = 20$ Links, tied to Γ by the proton
$\cos \omega = \text{num}_b / \text{den}_b$ matter, the scalar; a quantum that leaves a body is a message of matter	[m , den] ([4000, 6000])	nothing	dimension 1, one real line; a charged body of the same pair is a second row of dimension 2, a plane	the holders of the content, $x = (L_b + L_g) / \Gamma$; the plane also the holder of the sign	m , the mass

the form with the tensions, or the Wronskian, with the field’s divisor E_s and, for the massless field, its rest. Its rotation at rest, its band, its coefficients and its wall, $w = 6 \text{ den } \Gamma^2$, follow from Eqs. (1) and (4); its parts are its sources’ count, one real line per source; its reach is set by its gap (Section 6.4); its reads follow from its dimension (Section 3.4), and it sources what it reads at the weight it reads with. Every record clicks, a field’s included, since no row declares whether it clicks. Table 3 gives the rows of the universe, and Section 6.6 the numbers of the world [1, a family’s declaration; the families from the rule][2, how to add a family].

6.1 The exact bands, and the third band

The line of one Node at rest, $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$, is exact in integers only where $2 \cos \omega_0$ is an integer. So the rule carries exactly three exact massive rotations, $2 \cos \omega_0 = 1, 0$ and -1 , with the periods 6, 4 and 3 and the pairs [1, 2], [0, 1] and $[-1, 2]$ (THEOREM, GAMEBOARD). A record on an exact band at rest in the vacuum leaves no remainder in its rest rotation, so it never clicks and only turns. The exact bands are exact only in the vacuum: at a Node of pace p_0 the band’s rest rotation is $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2$, an integer at $p_0 = \Gamma$ alone, so inside a well a record on an exact band leaves a remainder and clicks like every family’s [1, the exact bands; the exact bands are exact only in the vacuum].

The bands [0, 1] and $[-1, 2]$ are allowed by the rule and unused by the universe [1, the third and the colours, closed as absent].

6.2 The rule’s own universe

The universe of the model is the one whose every number is the rule’s own: the pairs from the integers 1, 2 and 3 of the three exact massive rotations and the three axes, every divisor 1 except gravity’s, and the remaining free numbers named. Its rows stand in Table 3, as the file `examples/events/rule.json` holds them (DERIVED from the two keys of each row) [1, the families from the rule; the rule’s own universe][2, the input].

The universe’s integers are the Node clock $\Gamma = 6,000$, a multiple of 6 so that every pair is exact over it; the quantum’s norm $T = 2^{15}$, the normalisation of the quadratic form; and the integers’ width, 63 bits, a bound and not physics. Γ and T are units, as c and $\hbar$ are in nature, and no line of Rule3 fixes T , since the rule is linear and the scale of its form is free. Matter reads gravity and the binding field plainly, $x = (L_b + L_g) / \Gamma$; charged matter, a

second row of the matter pair at the dimension 2, reads the holder of the sign as well; light reads the same holders of the content and is bent by them, and does not read the holder of the sign, its own field. The bound charge's field, the exact band [1, 2] at under one Link's reach, was read by neither of the implementation's two worlds and has left the universe file (ASSUMPTION) [3, nothing that is not Rule3 remains]. Every field of the file is a record stepped by Rule3 at its pair and written by the one write; no field is laid by hand during a run, and a world's starting rest is the fixed point of the division act, computed once before the first interval (ASSUMPTION) [1, the final unification][3, two units and the bounds; nothing that is not Rule3 remains].

6.3 What each family carries, and how gravity enters as a clock

What a family carries is set by what Rule3 reads and what a record can write, not by a name (DERIVED, GAMEBOARD). Rule3 reads two things of a Node and no third: the clock's pace p_0 , one number, and the Link's paces (p_x, p_y, p_z), the three directions of the rule's vector. So a field that acts on the paces carries at most one time part and three axis parts, $1 + 3$. A record of one real line, neutral matter or light, has a form D and a stress and no Wronskian; a record of a plane, charged matter, has a Wronskian W as well; a field is always real, one line per source, since a plane's own W would be its own source. The shapes follow with no choice: *the scalar*, one real line, for matter, light and the binding field; *the plane* for charged matter; *one plus three*, the time part with the three tensions, for gravity, sourced by the form and by the diagonal of the record's own stress (Section 4.3); and *two real lines never summed at a Node* for the pair family of light in Bell's world. The law as it stands takes no vector part of a field and no off-diagonal stress, since Rule3 reads them nowhere; for a real record no current's read exists within the three tests, so no frame dragging of neutral bodies (Section 10.3) [1, the final unification; the tension][3, every family has a dimension].

How gravity enters: a slowed clock, and no curved space. There is no metric, no force and no law of motion in the model. Gravity enters in three steps. (i) *The write*: at every Node in each interval every family of quanta writes its form over the action, $D_i \text{ div } T$, into gravity's field over the divisor E_g , by the second line of Eq. (1). (ii) *The field*: gravity's field, [1, 1], steps by Rule3 at the pace 1, so around a static source it rests at Poisson's solution, $3G(r)s$, which is $3s/(4\pi r)$ far away; a change travels at one Link per interval, retarded, and the exact inverse is kept. (iii) *The read*: every family reads the field's level c at its Node into its own coefficients, Eq. (3), the clock's pace once and each Link's pace twice. What nature calls curved space is here the Link's pace, light's index $n = e^{2x}$; no space is bent. The factor two on the Link is the one assumption of Section 3.4, the clock's composition the rule's own [1, the paces; the clock once and the Link twice][1, the same clock and the same ruler for every act; the paces compose].

6.4 The reach of a field, and the binding field

A field's rest solves $\text{num } \Delta^2 a - 6(\text{den} - \text{num}) a = -3 \text{den } \sigma$, its line with the source once an interval, with Δ^2 the six-neighbour second difference. Outside a body its rest therefore falls by $e^{-\kappa}$ per Link with $\cosh \kappa = 3 \text{den}/\text{num} - 2$, the family's own band at zero rotation: the gap is both the mass of the field's quanta and the inverse reach of the field, one number (DERIVED, GAMEBOARD). [1, 1] has no gap, and its rest is Poisson's over the whole GameBoard. [1, 2] has $\cosh \kappa = 4$ and falls by 0.127 per Link, a contact well of $1/\kappa = 0.48$ Links' reach. A pair with num just under den has $\kappa^2 = 6 \text{den}/\text{num} - 6$ to second order, a range $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ Links, which is Yukawa's form [42] (S.11) [1, the well].

Assumption 4 (the binding field and gravity's divisor, decided with a reservation). The well of a record, $D_i \text{ div } T$, is laid into two fields. The binding field, a family with a gapped pair $[\text{num}_b, \text{den}_b]$, $\text{num}_b < \text{den}_b$, at the divisor 1, stepped by Rule3 at its pair as the massless field is and started at its rest, which with the gap is a sum with a range, $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$ Links and $e^{-r/R}/r$ beyond it: it binds the bodies, it is the model's strong force; its own travelling events have the gap $\cos \omega = \text{num}_b/\text{den}_b$, a quantum with a mass whose Compton length is R , Yukawa's own line [42] and no input. The massless field [1, 1] at the divisor E_g declared by the file: gravity, long-ranged and weak by E_g . Every family reads both levels into its pace at the weight 1, the two fields among the readers, each its own level among the content (the law's line, Section 3.4).

The first pair is [2400, 2401], with $R = 20$ Links, and the field's quantum has the rest frequency 0.029 per interval. The pair is tied to Γ by the proton: a cloud of 1,836 quanta is bound under $R = 20$ and not under $R = 10$ or 15 (Section 7.3). In worlds with bodies E_g is declared large enough that gravity does not enter a body's binding, 1,000 in the computations of Section 7, a world's own number under the bound of Section 10.2. Nature's E_g is about

3×10^{42} , and the hierarchy of gravity and electricity is the file's one number, as G is nature's. The decision carries a reservation, that it may be wrong [3, the two rows][1, what is open, item 20].

6.5 The vacuum content

Assumption 5 (the vacuum content is 0, a declaration fixed by nature's clicks). The massless field rests at a declared level everywhere, the vacuum content, the key **rest** of the universe file, derived by no line of the law; declared 0 in the file's integers at $\Gamma = 6,000$.

A uniform rest c_{vac} is a change of units (Section 3.4), which only nature's clicks can tell. Under the law's line every field reads the content as every family does, so the massless field's own travelling events have light's paces identically in every level: gravity's events and light through the same wells arrive together, the difference of their two Shapiro delays 0 and $\gamma_{\text{GW}} = \gamma_{\text{EM}}$, where nature's GW170817 bounds the difference of the two delays below 1.2×10^{-6} of light's [43] (DERIVED from the law's line; CLICKS; S.12). Inside matter the source is read through the pace, gravity gravitating (DERIVED, GAMEBOARD). With the two speeds one speed the measurement bounds no rest, and the vacuum content stays a declaration, 0 in the file and, under the vacuum's write of Section 7.5, the faces' positive content $\Lambda R^2/6$ (ASSUMPTION). [1, every row reads the content; what is open, item 22][3, the vacuum content is 0, a declaration fixed by nature's clicks].

6.6 The world's numbers, and how the universe was chosen

The universe is chosen by one principle stated before any row: one rule closes everything; the families are its inputs, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the rule allows (ASSUMPTION) [3, the standing goal; not one number in the engine, and as few in the files as the law allows]. The road to them:

(i) a family is a band of the rule, and its row holds only its two keys; (ii) three exact massive rotations, the pairs [1, 2], [0, 1] and [-1, 2] (THEOREM, Section 6.1); (iii) the three of Rule3 is the three axes, so the massless holder of the content carries 1 + 3 parts and every other family is a scalar; (iv) one denominator, every pair written over Γ , a multiple of 6; (v) the pairs from 1, 2 and 3 (ASSUMPTION): [1, 1] for gravity and the charge's field, and [2, 3], the first pair past the exact bands, for matter, nature's pair far past it (Sections 8 and 10.1); (vi) every divisor 1, under which the universe carried no free number but the mass; (vii) two numbers where nature's scale forced them, the binding field's gapped pair and gravity's divisor E_g (ASSUMPTION, with the reservation of Section 6.4); (viii) the vacuum content 0 where no write is declared (Section 6.5).

What remains: two units, Γ and T ; one bound, the integers' width; and four numbers of physics, matter's numerator m , the binding field's pair, gravity's divisor E_g and the vacuum content, and under the hypothesis of Section 7.5 one more, the vacuum write's rate, with every other divisor 1. The two worlds read a light-only copy of the file with the two holders of the content at the divisor 10^6 , so that light's own well stays under one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION); the electron's mass over Planck's and the proton's mass over the electron's are open and not derived (Section 10.3) [1, the rule's own universe; the final unification][3, the rule's own universe is the universe; the final unification].

7 Bodies

7.1 What a body is, and how it is computed

A body is its record over its Nodes and the quanta that record carries, and never one Node. Its count is the share of its record (Section 4.4), a reading and not a state; its form $D_i \text{ div } T$ is written into the fields it sources, and the fields enter Rule3 through the paces alone. The record is bound by the wells its form sources, by the binding field's screened sum with the range R , while gravity's $1/r$ is weak by E_g and enters no body's binding. No law is written for a body: it is the support of a standing mode of the one rule, and its record moves by Rule3 alone. Four statements make a body (DERIVED from the line, with the decisions named): (a) the body is the fixed point of its binding field, standing where the sources return themselves; (b) a field's level is the rest the start solves from the bodies' forms, after which Rule3 steps it; (c) a body's quanta are its record's share, carried wherever its record's current carries it; (d) a free packet or a cloud falls as $a = \Delta |d\omega_b/dc| \nabla c$, and a compact body by its tail alone, more slowly [1, what a body is; the four lines of the body][3, what a body is].

How a body is computed. By the rule's own iteration and no other formula: start from a spread of the declared count over the Nodes; write the wells that spread lays into the fields and let them rest; find the record's top mode in

those paces by Rule3's own read act, iterated; scale its two levels so that the share carries the declared count; write the wells that record lays; and repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: a wide start reaches the extended body, the cloud, and a narrow one the compact pixel, two fixed points at one count. [1, the generator; the four lines of the body][3, the generator is Rule3].

7.2 Stability

Let a body of M quanta have the record ϕ over its region, $\sum \phi_i^2 = 1$, with the counts $c_i = M\phi_i^2$. Let every field r it sources be a hollow ($\sigma_r = +1$) or a hill ($\sigma_r = -1$), with the static kernel $\mathbf{G}_r$ of its line and the divisor E_r , so that the well at a Node is $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$. To first order in Φ/Γ the well raises the one-Node rotation $2\cos\omega$ by $4g\Phi/\Gamma$, $g = (\text{den} - \text{num})/\text{den}$, and the body's fixed point is a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3\text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once because the record is its own source (Hartree's form). The record is the top mode of Rule3's symmetric form at the paces, and $dF/d\phi = 2\lambda\phi$ on the sphere is the fixed point's own equation with $2\cos\omega_b = 2\cos\omega_0 - \lambda$ (THEOREM, GAMEBOARD; S.13). Let $\phi_R(x) = R^{-d/2}\phi(x/R)$ be the body dilated, with d the GameBoard's dimension; the pressure is K/R^2 and each field's well energy is W_r/R^{s_r} , with s_r the field's scaling power at the body's size: $s_r = d$ beyond the field's reach and $s_r = d - 2$ within it or for the massless field [1, the functional and the dilation].

Theorem 4 (the stable body, under the dilation and to first order in the well). *At a fixed point of the binding row of width R the virial $2K = \sum_r \sigma_r s_r W_r$ holds, and the second variation of F along the dilation is $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$. A body is stable only where this sum is positive; it is a minimum of F where further the Hessian on the modes orthogonal to the dilation and the translations is positive.*

Proof. $F(R) = K/R^2 - \sum_r \sigma_r W_r R^{-s_r}$; $F'(1) = 0$ is $-2K + \sum_r \sigma_r s_r W_r = 0$; $F''(1) = 6K - \sum_r \sigma_r s_r (s_r + 1) W_r = \sum_r \sigma_r (3s_r - s_r^2 - s_r) W_r$: Derrick's scaling [44], Pohozaev's identity [45] on the GameBoard [1, the functional and the dilation]. $\square$

The theorem is a statement about the functional of Eq. (12), to the first order in the well, along the one direction of the dilation, with the dilation taken as continuous; it is not a theorem of the integer step on the full lattice, and the Hessian beyond the dilation is open (Section 10.3).

The factor $s(2 - s)$ is $+1$ at $s = 1$, 0 at $s = 2$ and -3 at $s = 3$: a hollow holds the body where its well falls more slowly than the pressure and breaks it where it falls faster. Three consequences follow. (a) One hollow beyond its reach binds every mass on a chain and in a box has one critical point, a saddle: the body disperses beyond it and collapses within it. (b) A massless field holds the body: Poisson's well in a box has $s = 1$, so $F = a/R^2 - c/R$ has one critical point, $R = 2a/c$, a minimum for every mass, which is Pekar's body [46]. Beside it a hollow beyond its reach adds a barrier, $F = a/R^2 - b/R^3 - c/R$ on a Gaussian body, with the wide critical point a minimum and the narrow one a saddle, and none where $3bc > a^2$, the collapse (COMPUTED, GAMEBOARD). (c) A collapse ends in a frozen clock: beyond first order the well saturates and the Link's pace weakens the pressure, both falling together as the clock composes, and since no pace ever reaches 0 a collapse ends in a cluster of Nodes whose clocks are frozen to the unit and never in a closed Link [1, the functional and the dilation][1, what is open, item 29].

7.3 The two branches: the cloud and the compact pixel

The fixed point of the binding field has two branches at every count above an edge, and the body of nature is the wide one (DERIVED from the fixed point and computed, GAMEBOARD; the identification is a decision of the model) [1, the two branches][3, the body of nature is the cloud].

The cloud, the body of nature. The wide branch is the continuum's Pekar body [46]: the count is spread over many Nodes, with the Pekar radius $5.25\Gamma E_g/C$ Links for a count C in the continuum's scaling, within a tenth of the table's. Its three masses agree within a few percent: the differences go as $(C/\Gamma)^2$ for a cloud free of the box and linearly for a cloud the box limits (COMPUTED, GAMEBOARD): the well it sources, its inertia from the momentum reading and its weight from the force term in a content gradient. Table 4 gives the three along the branch (an open 41-cube, $\Gamma = 6,000$, the counts in the vacuum's share) [1, the rows against nature, (u)].

Table 4 The cloud, the wide branch of the binding row's fixed point, along its branch (DERIVED from the fixed point, a reading of the algebra and no click; GAMEBOARD): the total count, the rotation $2 \cos \omega_b$ against the band's top 1.3333, the rms radius in Links, the count at the centre Node, the well and the inertia over the count and over each other, and the fall against a free packet's. An open 41-cube at $\Gamma = 6,000$, matter [4000, 6000], $T = 2^{15}$, under the massless field alone at gravity's divisor $E_g = 1$, Poisson's universe and not Table 3's; the counts in the vacuum's share at the Node.

Total count	$2 \cos \omega_b$ (top 1.3333)	rms radius	Centre Node	Well / count	Inertia / count	Inertia / well	Fall / free
4,000	1.3509	8.7	2.6	0.985	1.061	1.077	0.92
3,000	1.3437	10.3	1.0	0.991	1.037	1.047	0.95
2,000	1.3383	11.5	0.4	0.995	1.022	1.027	0.97
1,500	1.3359	12.0	0.3	0.996	1.016	1.020	0.98

So the universality of free fall [35] holds in the model for extended bodies to 2 to 8 percent in the computed regime, quadratically in C/Γ for a free cloud and linearly for a box-limited one, and the better the larger Γ is against the body's count.

The electron and the proton. The electron is one quantum of the matter family laid with a sense, a message and not a body, stable because one quantum does not disperse into fractions. The proton is a cloud of 1,836 quanta, so $m_p/m_e = 1,836$ is a ratio of counts and fixes no Γ ; the model does not derive it, and the proton's count is an input, declared in the world file as every body's count is (ASSUMPTION). A body's mass is a click mass: its inertia is read by its click period and de Broglie's k , its weight by the fall's clicks, its well by another body's orbit and its rest energy by the sum line's photons; nature's proton mass is such a click mass, its quarks' rest no part of the comparison [1, the three masses of a body][3, only clicks against clicks] [1, the electron needs no band of its own; the rows against nature, (t)].

The lightest body, and the binding range. Under the binding field a cloud stands only within the range R : the field holds no cloud whose Pekar radius $5.25 \Gamma/C$ Links exceeds the range, so there is a lightest body, $5.25 \Gamma/R$ quanta, 3,150, 2,100 and 1,575 at $R = 10, 15$ and 20 with $\Gamma = 6,000$, the lightest baryon, and the electron is never a body. The cloud of 1,836 is not bound under $R = 10$ or 15 and is bound under [2400, 2401], $R = 20$, its well 0.997 and its inertia 1.011 of its count and its fall 0.988 of a free packet's. So the file's pair is [2400, 2401], and the range is fixed by the proton (COMPUTED, GAMEBOARD) [1, the rows against nature, (w)].

7.4 A body's clicks

What a detector reads of a body follows from the body's record at its Nodes, the readings of Section 4.3 and the paces, with no further line; every formula below is a derivation with its status and its fence, and none is a run's result [3, the mathematics of the bound bodies and how it bears on the clicks].

A steady rotation writes no light. A body at rest with the plane record $\text{re} = A \cos \omega_b t$ and $\text{im} = eA \sin \omega_b t$, e its ellipticity, has the Wronskian $W = -eA^2 \sin \omega_b$ at every interval, so the write of the holder of the sign is a constant source whose field stands at its rest: no travelling events, no click anywhere. A body in uniform motion radiates nothing either, since its retarded level follows it. A breathing body, $A(t) = A_0(1 + \delta \cos \Omega t)$, has the Wronskian $-eA_t A_{t-1} \sin \omega_b$, which to the first order in δ and for a slow breathing, $\Omega \ll 1$, is $-eA_0^2 \sin \omega_b (1 + 2\delta \cos \Omega t)$, so the light born carries the breathing's frequency Ω and not the rotation ω_b , with $2\delta eA_0^2 \sin \omega_b / T$ quanta per interval in its first harmonic, and a far detector's click rate is that current's share through its front. So a symmetric body at rest clicks nowhere, and a body is seen by what it absorbs and by what it writes when it breathes or moves (DERIVED; the Wronskian and the source GAMEBOARD, the far detector's rate CLICKS) [1, light is born by the write].

The invariant, and the count's drift. For one Node of a body, $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$ with ω changing slowly has the solution $a_t = C \cos(\sum \omega + \phi) / \sqrt{\sin \omega(t)}$, so $A^2 \sin \omega_b$ is invariant, the discrete adiabatic invariant of Rule3's line, while the form $D = A^2 \sin^2 \omega_b$ and the share are not; the recurrence stepped over 20,000 intervals keeps $D/\sin \omega$ to 4×10^{-5} . The share in quanta at a Node, $(3\Gamma^2 / \sum_a p_a^2) A^2 \sin^2 \omega_b / T$ (Section 4.4), therefore moves under a moving well as

$$\frac{N(t_2)}{N(t_1)} = \frac{\sin \omega_b(t_2)}{\sin \omega_b(t_1)} \cdot \frac{\sum_a p_a^2(t_1)}{\sum_a p_a^2(t_2)}, \quad (13)$$

with no current through any Link: the first factor is the rotation's own change under the clock's pace, the second the Link paces' weights in the share (DERIVED for the invariant, to the first order in the slow variation, and for the drift; GAMEBOARD; S.14). The invariant is what nature calls the number of quanta, E/ω , and the share is nature's

energy; the same identity is Planck's relation on the lattice, $\text{share} = I \sin \omega$ with $I = A^2 \sin \omega / T$, so that $E = \hbar \omega$ holds with T for $\hbar$ and $\sin \omega$ in place of ω . The engine's declared region credits the share, so it credits two beams of equal power the same number of quanta; a body's click is one transition, I per click, under the detector's declared whole below (DERIVED; the whole ASSUMPTION; CLICKS) [1, the count is the record's share; the paces' own change moves the share with no current].

The clock's click rate. A bound body's rotation reads the content once: at rest $\cos \omega_0(U) = \cos \omega_0 + 2gU$ to first order, with $g = (\text{den} - \text{num})/\text{den}$, so its rate falls by $f(\omega_0)U$ with $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, 1.063 at [2, 3] and 1 as the gap closes; the rate $1 - U$ its small-gap limit. A light clock between two faces falls by $2U$, the Link's pace read twice (Section 10.2; S.15) (DERIVED, CLICKS).

The resonance. A bound record at the rotation ω_b reads a light level $b_L \cos \omega_L t$ into its content at the weight 1, the content x_0 at its Node otherwise static. At $k = 0$ the Node's line is $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$ with $2 \cos \omega = 2 - 2g(1 - 2x + 2x^2)$, so the light modulates the rotation by $\epsilon = 4g(1 - 2x_0) b_L / \Gamma$. The resonant part of the modulation at $\omega_L = 2\omega_b$ grows the amplitude at $\mu = \epsilon / (4 \sin \omega_b)$ per interval, the share at 2μ , inside the full width $\epsilon / \sin \omega_b$ (S.16). This is the one-mode case; the same modulation mixes the record's frequencies, so a body with two bound modes $\omega_i < \omega_j$ is driven at the sum $\omega_i + \omega_j$ and transfers between the two at the difference $\omega_j - \omega_i$, the line between two bound states, at the rate $\Omega_R = \epsilon_{ij} / (4 \sqrt{\sin \omega_i \sin \omega_j})$ for a modulation ϵ_{ij} of their coupling, Rabi's line (DERIVED, two derivations) [1, the two-mode line]. The read's side is derived; the fraction absorbed, with the body's write back, is open (DERIVED for the resonance, COMPUTED for the numbers; S.16; the record under the light GAMEBOARD, the body's share rising at its front CLICKS). Beside it: the one-mode line at $2\omega_b$ is the parametric resonance of Mathieu's equation; the two-mode line at $\omega_j - \omega_i$ is the atomic transition's analogue, Rabi's power-broadened line [1, the rows against nature, (y)].

A body's count in clicks. A body's count is its share summed over its Nodes, what a full absorber credits, and under a moving well it drifts by Eq. (13) (DERIVED, CLICKS). The whole of a click is the detector's. The engine's declared region credits the share, a bolometer's reading: two beams of equal power are credited alike, Millikan's slope 0. Nature's detector is a body, and the model declares its click as one quantum's change in the absorbing body, the detector's fourth declaration beside its window, its region and its floor (ASSUMPTION) [3, the whole of a click is the detector's own transition]: the absorber answers at its own rotation by the two-mode line above, one transition per click, taking the share $T(\sin \omega_j - \sin \omega_i)$, which is one photon's share $T \sin \omega_L$ to the order $\omega_i \omega_j$, below 10^{-18} at nature's gap. So a body counts photons and reads $\hbar \omega$ per click, Millikan's slope T , Planck's relation above with T for $\hbar$ (DERIVED, two hands; S.14); through a deepening well the number of quanta is kept, as nature keeps it, where the share falls 4.0 percent with I kept to 10^{-5} , the two counts agreeing in every static world to 10^{-15} . The share stays the conserved form and the draw's measure, Born's rule, so every ratio of the gates is unchanged and the gates are blind to the whole (THEOREM); the two slits' absolute N is read with the detector's own quantum as the whole, reported beside the whole the gate used. The engine's regions credit the share today, the detector's whole its next round (CLICKS) [1, the count is the record's share]. The formulas of clicks stand together in Table 2 with their statuses.

7.5 Clusters: how they enter, and what each thing represents

How a cluster enters. The model writes no line for a cluster. One assumption names the universe it describes: every body is bound, no body is free, and a record that is in no body is a message between two clicks (ASSUMPTION). A cluster is then bodies standing in one another's wells, bound by the same three fields and clicking at one another; between two bodies there is nothing but the click, the light of the holder of the sign and the two real fields. A Node is a body's Node or vacuum (DERIVED, GAMEBOARD) [1, the bound body is one Node; the universe is bound; the algebra of clusters].

The clicks between bodies are clicks in the one sense. A body is a region of Nodes with a front, the Ports where its tail meets its neighbour's, so a body is an instrument in the meeting's form: the click between two bodies is the pair (R, t) , one whole quantum credited to the region of the taker at the interval t , drawn by the share Q_R / W_c of the inflow through its front on the forward orbit. An exchange, a join, a parting and a message taken are runs of such clicks; a body's count is its share summed over its Nodes, and it moves only by the clicks at its front, up to the drift of Eq. (13) (DERIVED from the definitions; CLICKS) [3, the click's algebraic definition; the click is the meeting of the future with the past].

What each thing represents in a cluster. (i) *The three levels at a Node are the nesting.* The binding field's level at a Node is the matter's tail, the level of the holder of the sign is the charge's well and the massless field's

level is gravity's well: the three kinds of binding by rank; a reading of the three names the clusters the Node belongs to, a diagnostic and no click (DERIVED, GAMEBOARD). (ii) *Three kinds of binding, not three levels.* Quanta bind into a body by the binding field's screened well, bodies into a charged cluster by the field of the holder of the sign, and charged clusters into a gravitational cluster by gravity's $1/r$ well; the third kind nests without bound. (iii) *The hierarchy is recursive.* A cluster of clusters is the same structure one level up; no number bounds the depth but the frozen clock at a Node (Section 3.4). (iv) *The click is the pointer,* the only thing that passes between a cluster and its neighbour, so the observer is a cluster and knows the world beyond its tail by its clicks (DERIVED, CLICKS). (v) *The couplings are at the edge.* Every divisor is 1 but gravity's E_g , the universe of clusters carries Γ and T and nothing else, and a stone is clusters of clouds (ASSUMPTION for the divisors, DERIVED for the rest; GAMEBOARD) [1, the levels at a Node are the nesting; three kinds of binding, not three levels; the hierarchy is recursive; the click is the pointer; the observer is a cluster; the couplings are at the edge][3, what a body is; the rule's own universe is the universe].

The clusters' cosmology. The universe of the model is a finite ball of bodies flying through the static lattice, every body pulled by the others' inverse square at Kepler's G_K (Section 8, (j)). A shell of dust at the radius a feels the mass inside it, so $\ddot{a} = -(4\pi G_K/3)\rho a$ and $\dot{a}^2/a^2 = (8\pi G_K/3)\rho - \kappa/a^2$, Friedmann's dust equations exactly for the ball's interior [47], $\kappa > 0$ where the ball is bound; Hubble's law follows, and a receding emitter's redshift is its period read at the resting detector times its own tick factor, $1+z = (1+\beta)/\sqrt{1-\beta^2}$ at $c_m = c$, nature's relativistic form to the second order, the ball's own clock shift beside it at the second order (DERIVED, two derivations; CLICKS; S.29). Dust alone decelerates, $q_0 = \Omega_m/2 > 0$, where the supernovae read $q_0 = -0.55 \pm 0.1$ [48][49][50], and no line of the rule as it stands turns the sign: a uniform level pulls nothing, a drifting Γ is excluded by the atomic clocks and lunar ranging at 2×10^6 and 350 times the rate it needs, and more matter decelerates more. One line writes a level without a body: a uniform write of the vacuum's row, $-q_{\text{vac}}$ into the content at every Node each interval over a wall E_v , whose static rest is a bowl, $U(r) = U_{\text{faces}} - |q_{\text{vac}}|(R^2 - r^2)/(2E_v\Gamma LW)$, the level rising outward and every body falling outward: Newton's cosmological constant, under which the ball's equations are Friedmann's dust-plus- Λ exactly, $\ddot{a}/a = -(4\pi G_K/3)\rho + \Lambda c^2/3$, $q_0 = \Omega_m/2 - \Omega_\Lambda = -0.55$ at $\Omega_\Lambda = 0.7$, the turn from deceleration to acceleration at $z = 0.67$. Its one number is a declaration read from the supernovae's click as LW is read from G , one unit of well per Node per 10^{70} to 10^{73} intervals at nature's Λ . Light's stability needs every level non-negative, so the faces carry a positive content $\Lambda R^2/6$ in place of 0, the law's one addition to ΛCDM , a declaration no click reads; nothing local moves, the bowl's pull at one astronomical unit 2×10^{-22} of the Sun's. The finding is closed by a hypothesis by name with one declared number, not derived (HYPOTHESIS; CLICKS; S.29) [1, the vacuum writes][1, the clusters' cosmology].

8 The forms of nature the rule arrives at

Each form below follows from the rule's coefficients, read by the geometric optics of a packet's band, before any run; the form of nature beside it is on the comparison side and is never an input. The coefficients contain no metric, no Lorentz transformation, no potential and no force, only the paces of Section 3.4, the clock composing by the rule and the Link the clock twice by the one assumption. $U = c/\Gamma$ denotes the model's potential at the point named, the clock's share per level, and U_b its value at the impact parameter b [1, the rows against nature].

Two potentials. The forms are stated in two potentials. The model's own potential is the clock's share, $U = c/\Gamma$, the content x that the paces read; no instrument reads it. Every test of gravity takes GM from Kepler's law of the orbits of matter and c from light, so it reads $U_K = GM_K/(c^2 r)$. In the model a free packet of matter of the pair [num, den] falls in a content gradient at $a = [(1-\epsilon)/(3(1-\epsilon/2))]| \nabla x |$ with $\epsilon = (\text{den} - \text{num})/\text{den}$, exactly from its band, Eq. (4), which with $\cos \omega_0 = 1 - \epsilon$ and $c^2 = 1/3$ is $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] | \nabla x |$, so the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes.} \quad (14)$$

Each form below, lettered as the law's rows are, is written first in U , where the composed paces give $\gamma = \beta = 1$ exactly in the post-Newtonian words, the metric $N^2 = e^{-2U}$ and $h^2 = e^{2U}$, and the redshift's factor $f(\omega_0) = 1 + \alpha_U$, $\alpha_U = 0.063$ at [2, 3], and then in U_K , where the same algebra gives $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$ and $1 + \alpha_K = \tan \omega_0 / \omega_0$: 1.5, 1.25 and 1.329 at [2, 3], and 1 as the gap closes. The two columns are one algebra, and nature's tests read the second (DERIVED, CLICKS; S.18, S.19) [1, the rows against nature][3, general relativity in the law].

- (a) *The redshift* (DERIVED, CLICKS). Two clocks of one family at the levels c_1 and c_2 tick in the ratio $1 - (U_1 - U_2)$ to first order; the rest rotation of a bound record, the clock this row names, shifts by $f(\omega_0)U$ with $f(\omega_0) =$

$2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, 1.063 at [2, 3] and 1 as the gap closes. Against the Newtonian potential the orbits measure it is $z = (1 + \alpha_K) U_K$ with $1 + \alpha_K = \tan \omega_0/\omega_0$, 1.329 at [2, 3], where nature's clocks give $|\alpha| < 10^{-4}$ [35]. Beside it: Pound and Rebka's line [51][35].

- (b) *The bending of light* (DERIVED, CLICKS). At a Node of content x light's index is $n = e^{2x}$, $1 + 2x$ to first order, because the Link's pace reads the content twice. Far from a body its field is the massless field's rest, $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$, with σ_{tot} the body's wells per interval, and a beam passing at the closest distance b turns by the transverse gradient of the index integrated along the path:

$$\alpha = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl = \frac{3\sigma_{\text{tot}}}{2\pi\Gamma} \int_{-\infty}^{\infty} \frac{b dl}{(b^2 + l^2)^{3/2}} = \frac{3\sigma_{\text{tot}}}{\pi\Gamma b} = 4U_b, \quad U_b = \frac{3\sigma_{\text{tot}}}{4\pi\Gamma b}, \quad (15)$$

twice the angle $2U_b$ that a Newtonian pull of the same potential gives. The factor two is the ruler going with the clock, $hN = 1$, the one assumption of Section 3.4. Against the Newtonian potential the orbits measure the bending is $(1 + \gamma_K) 2GM_K/(c^2 b)$ with $\gamma_K = \text{den}/\text{num}$, 1.5 at [2, 3], where Cassini's tracking gives $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [35][52]. Beside it: Einstein's $4GM/(c^2 b)$ [35].

- (c) *The delay of light* (DERIVED, CLICKS; S.20). Light's coordinate speed at the content U is $e^{-2U}/\sqrt{3}$, so its index is $1 + 2U$ to first order, and a signal grazing a body at the impact parameter b from x_1 to x_2 , with $U = \mu/r$, is late by $2\sqrt{3}\mu \ln(4x_1 x_2/b^2)$ intervals: Shapiro's delay with $\gamma = 1$. Beside it: Shapiro's delay [53].
- (d) *The perihelion* (DERIVED, CLICKS). By the same geometric optics the matter pair's periapsis advances by $6\pi U_1/(a(1 - e^2))$ per orbit, Einstein's value, with the clock's second order; under the composed clock $\beta = \gamma = 1$ exactly, Mercury's 43 arcseconds per century. Against the Newtonian potential the orbits measure the advance is $(2 - \beta_K + 2\gamma_K)/3$ of Einstein's, 1.25 at [2, 3], where the planets' and the Moon's orbits give $|\beta - 1| < 10^{-4}$ [35]. Beside it: Einstein's $6\pi GM/(c^2 a(1 - e^2))$ [35].
- (e) *The moving clock: the band's own limiting speed and the fourth order* (DERIVED, CLICKS; S.21). A packet moving at the velocity v has its tick lengthened by the dispersion of its own band. The tick's factor is $f = \Omega(k)/\omega_0$, with $\Omega(k) = \omega(k) - k\omega'(k)$ the rotation at the moving centre. From Rule3's exact band of a free record, $\cos \omega = (\text{num}/(3 \text{den}))(2 + \cos k)$, the expansion in v gives $f = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0) v^4/(8c_m^4) + O(v^6)$ with $c_m^2 = \omega_0/(3 \tan \omega_0)$, the family's own limiting speed. The second order is the band's kinetic form at its own c_m and no test. The fourth order is $-(2\omega_0/\sin 2\omega_0) v^4/(8c_m^4)$, its coefficient 1.6926 at [2, 3] and 1 as the gap closes, the one number of this row. The limiting speed is $c_m^2/c^2 = \omega_0/\tan \omega_0 = 1 - \omega_0^2/3$, 13 percent below light's at the computed pair and within 10^{-27} of 1 at nature's matter pair (Section 10.1), where nature bounds the difference of matter's and light's limiting speeds at 10^{-15} to 10^{-20} [54][55]. Beside it: Lorentz's factor, reached as $c_m \rightarrow c$ [3, the click].
- (h) *De Broglie's relation* (DERIVED; GAMEBOARD for the band, CLICKS for the fringe spacing that reads it; S.22). A free record of a matter family has the band $\cos \omega_b = (\text{num}/(3 \text{den}))(2 + \cos k)$, the group velocity $v = (\text{num}/(3 \text{den})) \sin k / \sin \omega_b$, and to second order $k = \omega_b v / c_m^2$ with $c_m^2 = \omega_0/(3 \tan \omega_0)$, the family's own light speed. Beside it: $\lambda = h/p$ with the rest rotation as the mass.
- (i) *The fall* (DERIVED, CLICKS). In a pace gradient a free packet or a cloud accelerates at $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$ toward the well. So bodies with the same $\Delta d\omega_0/dc$ fall alike, and a cloud falls as a free quantum falls to 2 to 8 percent in the computed regime (Section 7.3, GAMEBOARD). Beside it: Galileo's fall and the equivalence principle [35], which hold in the model for extended bodies and not for collapsed matter.
- (j) *Newton's pull and orbits* (DERIVED, CLICKS). The massless field rests at Poisson's solution with the sources and at Laplace's outside a body, $c(r) = 3s/(4\pi r)$, so the acceleration falls as $1/r^2$ and a circular orbit's v^2 as $1/r$. Beside it: Newton's pull and Kepler's fall-off.
- (m) *A free record keeps its wave number* (THEOREM, GAMEBOARD). Rule3 is invariant under translation, so a free record keeps its wave number and rotation exactly, and a radiating body's period is read unchanged at every distance. Beside it: Hubble's law as Doppler clicks of matter flying apart through the static lattice, with no stretching of space, the vacuum's write adding Newton's cosmological constant by name (Section 7.5).
- (p) *The lens: a content bends light and matter alike, by one formula* (DERIVED, two derivations; GAMEBOARD for the index, CLICKS for the focus; S.28). A region of content is a lens for every family, as glass is for light: a record is slower inside it, so a window thick in the middle delays the middle more and the rays converge. A stationary boundary conserves the rotation, so a record of the pair $[\text{num}, \text{den}]$ entering a content U has its wave number inside fixed by $1 - \cos k_{\text{in}} = [1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}]/N^4$ with $N = e^{-U}$ the clock, exact in k where the right side lies in $[0, 2]$, the propagating domain. For light, $\text{num} = \text{den}$, the ratio is exact in the law's own variable, $\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$, so the phase index $k_{\text{in}}/k = (2/k) \arcsin[\sin(k/2)/N^2]$ is e^{2U} , $1/(1 - 2U)$ to first order, at long wavelength alone and carries the lattice's own chromatic term $k^2(e^{6U} - e^{2U})/24$ at finite k ; the bending, the delay and the lens are read in the ray limit $k \rightarrow 0$: the gravitational lens. For slow matter,

$k, k_{\text{in}} \ll 1$, the index is $n_m^2 = k_{\text{in}}^2/k^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num } k^2)]/N^4$, growing as $1/k^2$, so the slower the packet the more it bends and one slab of content focuses slow matter far more than light: the fall of a body is the refraction of its own record, Newton's well the $1/k^2$ term and the Link's double read on top, their ratio the Kepler column of Eq. (14). The lens acts on one quantum at a time: a packet of one quantum repeated M times credits the focal detector $M\Phi_f/W_c$ clicks. Beside it: nature's lens $n = 1 + 2\Phi/c^2$ for light and $n^2 = 1 + 2\Phi/v^2$ for a slow body, the law's two terms at a small gap; the model's own, to our knowledge, is the one exact lattice form for every family, with the pair's factor $6(\text{den} - \text{num})/\text{num}$ in place of the mass [1, the content's index for every family; the rows against nature, (p)][3, one quantum through a lens, a beam splitter and an interferometer is the law's own].

What goes in and what comes out. The model puts in one rule, the paces' read under the one assumption of Section 3.4 and the files' numbers. In the model's own potential the bending (b), the delay (c) and the perihelion (d) are Einstein's exactly, $\beta = \gamma = 1$, the geodetic precession 1.00 of Einstein's and Cassini's γ exact. At the rule's own matter pair the model therefore does not describe the gravity of the solar system; the closing gap is nature's pair, where $\gamma_K - 1$ is below 10^{-18} to 10^{-27} (Section 10.1), and whether the model's bodies, the binding range and the proton's cloud stand at that pair is not shown in this paper. Read as a metric, the composed clock gives $N^2 = e^{-2U}$ and the ruler $h^2 = e^{2U}$, against the isotropic Schwarzschild chart's $1 - 2U + 2U^2$ and $1 + 2U + 1.5U^2$, the spatial second order 2 against 1.5. One click of the strong field reads the difference: light's index e^{2U} has its photon sphere where $d(nr)/dr = 0$, at $r = 2m$ with $U = m/r$, so the capture radius of a black hole's shadow is $b = 2em = 5.44m$ against Schwarzschild's $3\sqrt{3}m = 5.20m$, 4.6 percent larger, a prediction by name, where the quadratic forms as written gave $8m$ (DERIVED, CLICKS; S.30). Beside it: the Event Horizon Telescope's shadows, Sgr A*'s fractional deviation from Schwarzschild -0.08 ± 0.09 [56], 1.4 standard deviations from the prediction, and M87*'s within 0.3 [57]. The force between moving charges follows under the holder's rotation from the lattice's own symmetry. Multiplying Rule3's step by the record's conjugate gives an exact continuity identity, so the sign's density wW and its halved Link current $-R_a J_a/2$ are one conserved four-current, the lattice's $j = \rho v$; and the time turn with the three Link phases, (θ_0, θ_a) , is one covector by minimal coupling, shifting (ω, k) as the band's boost does. So the odd lines are sourced by the halved current over the same wall den T as the time level, no number matched, and a co-moving reader feels Coulomb's force over γ , 0.995 at $0.1c$ and 0.866 at $0.5c$ (DERIVED, two hands; CLICKS). Its domain: the symmetry is the band's at long wavelength and breaks at the lattice's $k^2/12$; a source and a reader of one family see one light speed, the band's own $c_s^2 = \text{num}_s/(3\text{den}_s)$, under which Rule3's continuity identity makes $(wW, -R_a J_a/2)$ a four-vector; at light's c the same to 10^{-28} in nature; and it is no new physics, the model's part that a lattice rule yields the covariance and the normalisation by itself, its look a check of the booking and not a test [1, the odd line's source is the sign's own current; the rows against nature, (b2)].

9 The two slits and Bell: the one experiment and the gate, derived and read by the gates

The paper carries no experiment against nature: the results of the two experiments below are derived from the rule and read by the engine's gates (Section 3.7). The two slits is the one experiment, its fringe row fixed by the rule's line in real arithmetic and its arrival interval the lattice's own number; Bell's world is a gate and not an experiment: its number is fixed by the declared credit and the parts' determinism. Nothing in a world is built to bring the right result [3, the paper; one experiment and one gate; nothing in an experiment brings the right result by construction].

9.1 The two slits: the world

The emitter is a declared packet of light. Its wavelength is 8 Links along $+x$, $k = \pi/4$. Its levels are a raised-cosine packet of about ten quanta per Node, $b = 1,328$, over $x = 4$ to 16, flat across $y = 4$ to 44. The wall is the inner face at $x = 20$ with two gaps of three Nodes, $y = 17$ to 19 and 29 to 31, $d = 12$ Links apart. The GameBoard is flat, $45 \times 48 \times 1$ with the third axis folded, its y faces open and its x faces receding; the screen is the last column, $x = 44$, $L = 24$ Links from the wall. The screen is twelve declared detectors of four rows each; the run is 130 intervals, read over the intervals 45 to 130. A record uniform along the folded axis has the three-dimensional band at $k_z = 0$, so the flat GameBoard carries the world's band exactly (DERIVED, GAMEBOARD). On this world the law's line gives 278.0 quanta through the screen in the physical labels (277.8 in the engine's), the peak of the screen's inflow at the interval 68.0 against 61.8 on the same world with the lattice refined fourfold (the continuum's limit 61.6) (COMPUTED, CLICKS) [1, the rows against nature, (g)][3, the two slits; one experiment and one gate].

9.2 The two slits: what the model says it must give

Three readings of the one world, each written before any run. (a) *The fringe*. The far field's spacing is $\lambda L/d = 16$ Links; at $L = 2d$ the fringe is the near field's, with the minima at the rows 15 and 33 (DERIVED, CLICKS; S.23). Light's line at the vacuum's paces, stepped in real arithmetic on the world with its source face open gives 280.8 quanta through the screen over the intervals 45 to 120, per region as the script's table lists (COMPUTED, CLICKS; the script `paper/general_formula/two_slits_real_line.py`); the Huygens sum of the two gaps gives 273 with the same central fringe. (b) *The clicks*. The screen's shares rounded to whole quanta are the expectation, and the row drawn by the shares at the declared seed is the clicks, with the draw's scatter $\sqrt{Np(1-p)}$ per region; on the Huygens blind row at $N = 273$ the central maximum stands above its two minima by 5.7 times its scatter. (c) *The lattice's own number*, the inflow peak's 68.0 against 61.8 (Section 9.1), which the click line resolves (COMPUTED, CLICKS). [1, the rows against nature, (g)][3, the click is the meeting of the future with the past].

9.3 Bell: the world

In the model's words, entanglement is one event with two trees. The pair is laid as the pair family, two real lines of light never summed at a Node, with equal parts. Each side's detector, a region at least half a wavelength across, reports the two parts' signed level sums at its ports and knows nothing of angles. The reader applies the declared setting, a pair of integers (p, q) , with the port coefficients $e(+) = (p, q)$ and $e(-) = (-q, p)$, exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are (1, 0), (1, 1), (12, 5) and (5, 12), at 0, 45, 22.62 and 67.38 degrees. The credit is the one form of the click: the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (16)$$

with α_k and β_k the part k 's amplitude at the port on each side and $e_k(p)$ the port's integer coefficient on the part k , accumulated over the window on both members of the level pair. The joint share is never the square of a local sum, which one detector could form alone and which gives S at most 2. One draw is made per pair: the first side from its marginal and the partner from the conditional (ASSUMPTION, CLICKS) [3, Bell in nature's own layout].

9.4 Bell: what the model says it must give

Three layers, two from the algebra of the GameBoard and the third a declaration. (1) Rule3 gives each part's amplitudes at the ports: the two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, identical bit for bit at every Node and interval (THEOREM, GAMEBOARD). (2) The meeting with the linearity of Rule3 gives the joint amplitude: the realised click at A's port weights the parts by A's analyser, is read back to the root where the two parts were one event and forward to B, where B's analyser weights them, and the root sums the two branches, $\sum_k c_k e_k(a) e_k(b) = \cos(a - b)$ for equal parts $c = (1, 1)$ (DERIVED, CLICKS). (3) That the share is the density of the meeting, its square, is Born's rule for a pair [41]: nothing in Rule3 forces it, as quantum theory postulates it too; it is the least non-local credit that reaches nature's correlations with every quantum counted, and everything after it, the fractions, S , GHZ's M , is derived in the algebra (ASSUMPTION, FITTED; CLICKS). From the three, with the pair's common phase θ uniform over the pairs and A's + port at a and B's at b ,

$$\begin{aligned} \text{the meeting:} & \quad P(++) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\ \text{a local credit by the shares:} & \quad P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\ \text{a local credit by the sign:} & \quad E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\ \text{the credit by the contrast:} & \quad E = \cos 2(a - b) \text{ at the efficiency } 2/\pi, \quad S = 2\sqrt{2}, \end{aligned} \quad (17)$$

The second, third and fourth lines follow by averaging over the pair's phase θ (DERIVED, CLICKS; S.10). So $E(a, b) = \cos 2(a - b)$ under the meeting. For the declared pairs it is exact, Lagrange's identity of the reader on equal parts: $E = ((p_{APB} + q_{AQB})^2 - (p_{AQB} - q_{APB})^2) / ((p_A^2 + q_A^2)(p_B^2 + q_B^2))$, so at the four settings $E = 119/169, -119/169, 120/169$ and $120/169$ and $S = 478/169 = 2.8284$ exactly, an efficiency of 1 by construction and the marginal one half on each side whatever the partner's setting (COMPUTED, CLICKS; S.10). For unequal parts, r their products' ratio at the ports, the same credit gives $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ with $\rho = 2r/(1 + r^2)$ and $S = \sqrt{2}(1 + \rho)$ at the ideal angles (DERIVED; CLICKS). The maximum of S over the four settings for $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ is

$2\sqrt{1+\rho^2}$, Tsirelson's bound $2\sqrt{2}$ at $\rho = 1$ and never the algebraic 4, and the declared pairs' $478/169 = 2.82840$ sits below $2\sqrt{2} = 2.82843$ by the rational angles (DERIVED, CLICKS).

What the GameBoard gives without the declaration is the second and third lines of Eq. (17), two theorems about local credits on the same world (THEOREM, CLICKS): a credit of each side by the shares of its ports gives $S = \sqrt{2}$, and a credit by the sign of its larger port gives the sawtooth, whose sum is exactly 2, Bell's bound. Bell's value in this model therefore sits in the credit rule, which is a declaration, and it is no independent evidence that the rule accounts for entanglement. Nature beside the derivation: $S = 2.42 \pm 0.20$ in the loophole-free test of 2015 that measured the sum [30], and the two photon tests of that year [31][32]; 2.42 is what quantum theory predicts for that experiment's state fidelity, $2\sqrt{2}$ the ideal state's. The same credit on four lines gives the three-quantum GHZ correlation $E_3 = \cos 2(a+b+c)$, every pairwise E 0 and Mermin's $M = -4$, $|M| = 4$ where every local credit gives at most 2, derived by two hands and read by the engine's gate exactly, the blind first (DERIVED under the declared credit, CLICKS) [1, the GHZ gate]. In the world, the two parts of a beam are one record stepped twice, so the lattice enters J only through the common factor $|\alpha\beta|^2$ and E is the Lagrange identity of the ports' coefficients; the gate's run checks the normalisation and the parts' equality and tests no physics beyond the credit. No signal passes between the sides, since the detectors read locally and each side's marginal is one half; no settings are chosen by the GameBoard's own past, so a superdeterministic world [58] is not this one [1, the rows against nature, (h)][3, the click is the meeting of the future with the past; Bell in nature's own layout].

10 Discussion: the constants, the calibration and the findings

10.1 The constants

The causal bound is one Link per interval, and light's own speed at long wavelength is $1/\sqrt{3}$ Links per interval (Section 3.3). The quantum of action is one click, whose energy is the quantum's norm T ; the quantum of distance is the Link and that of time the interval. Newton's constant is not declared: a body whose wells are s quanta per interval makes the massless field's rest $3G(r)s$ at r Links over the field's divisor, $3s/(4\pi r)$ far away, and the level slows the clock by the potential $\Phi = -c/\Gamma = -U$, Newton's sign, so (DERIVED, GAMEBOARD; S.25)

$$\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi \Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi \Gamma E_g} \text{ per quantum of well, in Links and intervals,} \quad (18)$$

a reading of Γ , the file's E_g and the lattice. With $c^2 = 1/3$, $G_{\text{clock}} = 1/(4\pi\Gamma E_g)$ is the number per quantum of well that lowers the clock's share, and Kepler's G , read from the orbits of matter as clicks, is $[2 \cos \omega_0 / (1 + \cos \omega_0)] G_{\text{clock}}$ by Eq. (14), 0.8 of it at $[2, 3]$ (DERIVED, CLICKS). The vacuum content, the massless field's declared rest, is the model's cosmological constant, 0 by nature's clicks (Section 6.5) [1, the constants].

The fine-structure constant, the units and the matter pair. Under the holder's rotation (Section 5.5) one circular quantum of the plane, $z = Ae^{i\omega_0 t}$, has the count $2A^2 \sin^2 \omega_0 = T$ and the Wronskian $|W| = A^2 \sin \omega_0$, so it writes $W/T = 1/(2 \sin \omega_0)$ per interval into the holder of the sign; and with the holder's rest read as an angle, de Broglie's k and light's c the force's constant is $\alpha_{\text{law}} = (3\sqrt{3}/8\pi)/(\Gamma \sin \omega_0)$ (DERIVED, two derivations; CLICKS; S.26). Under the identification of the electron as the free quantum of matter, nature's $\alpha = 1/137.036$ reads $\Gamma \sin \omega_0 = 28.33$. With den the family's own integer the least pair $[\text{den} - 1, \text{den}]$ has $\text{den} = 2/\omega_0^2 = \Gamma^2/401.3$, so the two clicks fix the pair and Γ alone stays free. The units follow: one interval is $\omega_0 \hbar / (m_e c^2) = 1.288 \times 10^{-21} \text{ s} \times \omega_0$ and one Link $\sqrt{3} \omega_0 \bar{\lambda}_e = 6.69 \times 10^{-13} \text{ m} \times \omega_0$. Light's band ends at $\omega = \pi$ along the body diagonal and at $\arccos(1/3)$ along an axis, so the model has a highest photon energy, $1.605 \text{ MeV}/\omega_0$ along the diagonal and $0.629 \text{ MeV}/\omega_0$ along an axis; LHAASO's photons of 1.4 PeV arrived [59] with no orientation of the lattice known, so by the diagonal's ceiling $\omega_0 < 1.1 \times 10^{-9}$, $\text{den} > 10^{18}$, $\Gamma = 28.33/\omega_0 > 2 \times 10^{10}$ and $\gamma_K - 1 = \text{den}/\text{num} - 1$ is below 10^{-18} (S.27); the Link's bound of Section 10.2, (2), below 1.4×10^{-26} metres, is tighter: $\omega_0 < 2.1 \times 10^{-14}$, $\Gamma > 1.3 \times 10^{15}$, $\text{den} > 4 \times 10^{27}$ and $\gamma_K - 1 < 10^{-27}$ (DERIVED, two derivations; CLICKS; S.27). At such a Γ the wall exceeds 10^{54} ; the engine's worlds at $\Gamma = 6,000$ are the law at the lattice's own scale [1, the law's alpha is a reading of Gamma sin omega.0; the units are the electron's and the band's top is the highest photon energy].

10.2 Nature's numbers enter only as clicks

Every measured number of an experiment is first written as a click pattern on a declared detector against a declared clock. The families' pairs and the units Γ and T , what in nature is one Link and one interval, are fixed by that match and by nothing read off the GameBoard; the two ratios the model fixes without it are light's speed of $1/\sqrt{3}$

Links per interval and the causal bound of 1 (ASSUMPTION, CLICKS) [1, the reading; nature's numbers enter the board only as clicks]. Every formula of Section 8 is derived in three steps: the law's line in the GameBoard's words; each word traded for a detector's report of it (ω for a clock's rate, k for a fringe's spacing, $A^2 \sin^2 \omega/T$ for the count, U for the ratio of two clocks' rates); and what remains, a relation among clicks alone, set beside nature's [3, only clicks against clicks]. The calibration series:

- (1) *The two speeds.* Under the law's line gravity's own events and light have one speed in every level (Section 6.5), so GW170817's bounds [43] are met by the line and read no rest; the vacuum content is declared 0. Gravity's divisor is bounded below by light's own fringes, which never move with the intensity in nature (DERIVED, CLICKS).
- (2) *Light's dispersion on the lattice.* Along an axis the group speed falls as $1 - 0.083 k^2$. MAGIC's bound on a quadratic dependence of light's speed on its energy, $E_{QG,2}$ above about 6×10^{10} GeV [60], bounds $|\delta v/v|$ below about 4.5×10^{-16} at 1 TeV, so the Link is below about 1.4×10^{-26} metres, a first bound on the unit of Γ (COMPUTED, CLICKS; S.24).
- (3) *Matter's two slits.* A matter packet's fringe spacing and time of flight give its inertia m^* , and nature's $h/(mv)$ gives the matter pair.
- (4) *The two clocks.* Under the principle a body in a level is the flat body in local units, its size $1/h$ Links, and each mode of its record scales by N in the law's variable $2 \sin(\omega/2)$, exactly at long wavelength and to the lattice's $k^2/12$ at finite k ; the temporal beat of two modes, $\Delta' = \omega'_2 - \omega'_1$, the frequency a detector reads, scales by N to the relative correction $(1 - N^2)(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)/24$, nil at nature's gap (10^{-38}) and 1.4 percent at the engine's $\omega_b = 0.78$ and $U = 0.1$; a cavity of bodies, L/h Links apart, round-trips in $2L\sqrt{3}/N$, the atom's rate exactly, so cavity and atomic clocks shift alike, as nature's do to 10^{-5} [35]; a cavity of declared faces ticks at N^2 and is not nature's (DERIVED from the principle; CLICKS; S.31).

Of the numbers, $\Gamma \sin \omega_0$, the gap, gravity's divisor and the Link are fixed or bounded (Section 10.1) and the vacuum content is declared; until the rest are named, run and read, the model's numbers are its own and not nature's, and the two worlds of Section 9 show that the implementation works and are not these experiments.

Three predictions by name, untested. Three numbers follow from the law's lines where general relativity and quantum theory do not meet today, each a derivation entered as a hypothesis by name, read by no click of nature yet, with nothing claimed beyond the number (DERIVED, HYPOTHESIS; two derivations). (1) *A clock in two places at once* (CLICKS). A bound body with a transition Ω , laid as one record on two paths at the heights h_1 and h_2 for a time T , has its gravity act on that one record through the clock's share alone, its beat Ω scaling as $N = 1 - U$ (Section 3.4); so the two paths' internal states differ at the recombination by the phase $\Omega \Delta U T$ and the fringes' visibility falls as $|\cos(\Omega \Delta U T/2)|$: 0.294 rad and 0.989 for a strontium clock at one metre for one second, with the redshift's sign; the semiclassical expectation [61] has the same magnitude, and to our knowledge it is unmeasured. (2) *The gravity of light* (GAMEBOARD). A light packet's pull on a static body is its count's, $1 \times E/c^2$, where general relativity's pencil pulls at $2 \times E/c^2$ [62], a blind for a world. (3) *The black hole's quanta* (CLICKS for the temperature). A collapse ends in a frozen clock from which no travelling event leaves (Section 7.2) and the step is a bijection (Theorem 1); so the law's black hole radiates nothing, $T_H = 0$ against Hawking's 6.2×10^{-8} K for one solar mass [63], unmeasured; its entropy its frozen Nodes' levels, extensive [1, the rows against nature, (a2)].

10.3 Findings against nature and open questions

GameBoard readings, not findings. The rule's own universe at [2, 3], no finding against nature, which measures at its own pair (GAMEBOARD): matter's limiting speed is 13 percent below light's at the computed pair and within 10^{-27} of light's at nature's pair (Section 8, (e)); against the Newtonian potential the orbits measure, the post-Newtonian observables leave Einstein's by 25 to 33 percent at the rule's own matter pair, the geodetic precession 1.33 of Einstein's against Gravity Probe B's 0.9993 ± 0.0028 [64], closing with the gap; no number of the computed pair is nature's (Section 8). The proton is computed as a cloud of 1,836 quanta at least the electron's Compton length across, 386 fm in nature's units, where nature's proton is 0.84 fm: a reading of a computed cloud whose click, the form factor read by scattering, has not been run (GAMEBOARD). A second reading is a hypothesis by name to be run (HYPOTHESIS): the proton one event of a three-line family with its own gap, a second matter family, under which the proton's mass over the electron's is the ratio of two gaps and no longer a count, and the event is at least its own Compton length across, 0.21 fm, nature's 4.0 times it; the form factor is the one look that separates the two readings, and the model keeps the cloud until it runs [3, the proton's two readings]. The charged cloud is bound a little less than the neutral one and is heavier by about 10^{-5} , where nature's neutron is the heavier by 1.4×10^{-3} : a reading of two clouds of one count, whose clicks are the clouds' masses (GAMEBOARD) [1, the rows against nature, (v) and (z)].

Findings against nature. With the diagonal tension alone, the massless field's travelling events carry the breathing polarisation and the lattice-aligned + polarisation and no $\times$, where nature's gravitational waves carry + and $\times$ at every orientation (CLICKS); beside it, light's band is anisotropic in the phase by $k^2/36$ between an axis and the body diagonal, of order 10^{-30} at visible wavelengths, which no click reads, the one bound, the PeV photons' of Section 10.1, met (GAMEBOARD) [1, what is open, item 23]. The law has no current's read for real records, and gravity's readers are real records, so a neutral gyroscope precesses by 0 from a rotating body's motion, where Gravity Probe B measured 37.2 ± 7.2 milliarcseconds per year against general relativity's 39.2 [64] (DERIVED, CLICKS) [3, general relativity in the law; the current's read]. The three, the drag's 0, the missing $\times$ polarisation and the α_1 below, are one structural fact: the model's metric is diagonal in the lattice's frame, the content's read real and even and the sign's a phase on the Link, so magnetism is covariant and gravity is not; in the post-Newtonian words a metric with no g_{0i} in its preferred frame has $4\gamma + 4 + \alpha_1 = 0$, the preferred-frame parameter $\alpha_1 = -8$ at $\gamma = 1$ against nature's $|\alpha_1| < 4 \times 10^{-5}$ from the binary pulsar PSR J1738+0333 [65], the lattice the preferred frame; a finding with two hands (DERIVED, CLICKS) [1, what is open, item 15]. The ball of dust alone decelerates, $g_0 = \Omega_m/2 > 0$, against the supernovae's -0.55 ± 0.1 , a finding by name closed by the vacuum's write, a hypothesis with one declared number and not a derivation (Section 7.5). And the nuclear binding, the deuteron's 2.2 MeV against the model's 10^{-13} of it, a finding by name with two hands (the strong force below).

The lattice's frame, and its mend as a hypothesis. Why kinematic clicks hide the frame and gravitational clicks read it (DERIVED, two hands; CLICKS): at long wavelength every record's band is a Lorentz form, $\omega^2 = \omega_0^2 N^2 + c_m^2 k^2$ with the family's own c_m , so the moving clock's γ follows and no Michelson-Morley shift appears at the band's order; the content's level is a Poisson rest in the lattice's frame each interval, and its kernel carries no speed, so a mass moving through the lattice writes the same bowl with no g_{0a} and no g_{ab} [1, the lattice's frame is read by gravity alone, and its carrier is a pair of Links; what is open, item 15]. The mend, a hypothesis by name with no line written and no number (HYPOTHESIS): the six off-diagonal components want a carrier of a second kind, a level on a pair of Links at a Node, sourced by the mass current and the off-diagonal stress and read by every record, the real ones included, which a phase on the Link cannot be; the boosted static metric already gives $g_{0i} = -4Uv$, Einstein's, so no coefficient is free. Against it stands a theorem with two hands: no real, local, explicit, bijective lattice line on Rule3's clock carries g_{0a} ; the three ways tried and a fourth close as one, the explicit real mixed term losing the conserved form (Theorem 3), the time-symmetric form the one-Link radius, the Peierls phase the real records, and the cube's two sublattices stepping half an interval apart carrying g_{0a} but leaking the Link's pace into every gapped record's rest rotation, a clock in a well redshifted at 5/3 of general relativity's, excluded by Pound and Rebka [51] at 60 standard deviations. So the three are the model's structural finding on Rule3's clock, and the carrier stays a hypothesis by name only as an object Rule3 would step, never a change of the line: the alternating clock, a one-sided mixed term and the diagonal hopping that would carry the $\times$ polarisation, each a change of Rule3, are refused by the decision that the line and its clock are fixed (Section 3.6). It is not derived [3, nothing that is not Rule3 remains].

The strong force, by derivation. On the board, the strong force is two things under one name. (i) Confinement exists and is the click's. A record of three real lines with a rotation of the three on each Link, the colours as the three axes, is consistent algebra (two hands: the conserved form under the Link rotations a theorem, the three colour charges conserved covariantly, colourless bodies at $\sum W = 0$). Such a record is one event, and a click credits a whole quantum and reports a region (Section 5.3), never one line of a record: no free quark, the quark model's three the dimension's three, the colourless bodies a quantum with its opposite sense or three coplanar colour axes, and a hard click reads the lines' shares as the partons' fractions; a theorem of the click's one form, with no coupling and no number (THEOREM, two hands; CLICKS). The record's holder is written through the same far rest as the sign holder's, so its coupling at the Link is the charge's α , and Rule3's real records, their count conserved, have no vacuum polarisation and no running (two hands); so Bjorken scaling is exact on the board, where nature reads scaling violations of 10 to 20 percent across the decades of momentum transfer, and the three lines' shares sum to one, where nature's valence quarks carry about half the proton's momentum: two findings by name, one hand each (CLICKS). (ii) The nuclear binding does not exist on the board. Nucleons are separable events, and their binding, 2.2 MeV for the deuteron and 8 MeV per nucleon in iron, is an energy released when they join, which the conserved form forbids at fixed paces (Theorem 3): a body's click mass is $M \sin \omega_b$ and its mass defect at a kept count is the redshift of its own rotation by its self-content, $M\omega_0 x_{\text{self}}$, 10^{-13} of the deuteron's (DERIVED, two hands); the one binding row binds two clouds of 1,836 at $16/\Gamma$ relative, and the binding per quantum times a body's size in Links is bounded by $0.24CW/\Gamma$, C the count and W the write weight, so a nucleon of 10^{10} Links bound at order one needs W of 10^{21} to 10^{22} , no integer of the lattice (THEOREM, one hand; GAMEBOARD); the credit's quantum $T = 0.511$ MeV is no divisor of 2.2 MeV. Nuclear binding at nature's Γ is reached by no route of the model as it stands, field

or click, a finding by name (two hands; CLICKS), and the binding stays a hypothesis by name with no mechanism, the colours' group and the three-line record's charge open [1, what is open, item 1][3, is the strong force not events that do not split].

The weak force. The model has no electroweak theory; its weak interaction is the conversion, a click, a hypothesis by name [1, the conversion]. A holder's gap is its range and never its strength: a gapped holder's kernel is Yukawa's with the same constant at the Link as the massless one, the gap entering only as $e^{-r/R}$ beyond, so every rotation holder, massive or not, couples at α at the Link (DERIVED, two hands; GAMEBOARD), and no holder Rule3 steps has the coupling $1/(\Gamma \sin \omega_0) = 1/28.33$. That this number lies within a few percent of nature's strong coupling at the Link's scale and of its weak coupling at the W's scale, which nature itself holds 0.7 percent apart, is one coincidence at a few percent probability, recorded and used for nothing (two hands); no G_F follows from it. The reading of a three-line record as (ν, e) under one $O(3)$ has no fourth rotation, a real line carrying no phase, so no Z , where nature has the neutral current [66] and the Z [67]; and Rule3 commutes with the cube's 48 symmetries, the reflections included, so no record of the model has a handedness and parity is conserved, where nature's weak interaction violates it [68]: two findings by name (two hands; CLICKS) [3, the weak-force reading].

The three roots of the findings. Each finding above is the price of one of the model's three virtues, and none is a number's fault: no choice of Γ , the level weight, T or a pair moves any of them (two hands) [3, why the gaps]. (1) The click is the model's one quantum element, the records deterministic amplitudes and the Links' levels written, so no coupling runs: α is fixed at $1/137$ where nature reads $1/128.9$ at the Z 's mass [67], a finding by name (one hand). (2) One denominator Γ reads every level into the paces, so no read of a level binds a nucleus: the bound $0.24 CW/\Gamma$, and the click route closed by the conserved form (THEOREM). (3) The step is explicit, local and bijective and the content's read real and even in the lattice's frame, so no g_{0a} and no g_{ab} , seven of the metric's ten components (THEOREM, two hands); and the same step commutes with the cube's reflections, so no handedness: the click's exactness, the one integer and the exact inverse.

Open. An exactly conserved integer charge (Section 5.5); gravity's divisor E_g (EXPERIMENT); the matter pair's gap (Section 8); what fixes Γ itself; and a current's read for real records [1, what is open].

11 Conclusion

One line of integer arithmetic at every Node of a cubic lattice, with one remainder kept at the Node, is reversible to the bit, conserves a quadratic form with its current at fixed paces, for any field and any tensions, without the rounding, and carries the momentum flux as the wave's own stress. On that lattice there are only events that split, run forward from an emitter as the future and read backward from a click as the past, and a click is their meeting at a declared detector, the choice among the meetings being the one non-local element. Gravity is a level that slows a Node's clock; the clock composes by the rule, and under one assumption, that a Link is clocked twice by its reading Node's own clock, the model's static metric is the exponential metric, the weak-field forms are Einstein's with $\beta = \gamma = 1$ exactly, and the shadow's capture radius $2em$ is a prediction; against the potential that Kepler's orbits measure the matter band's factors reach 1 only as the family's gap closes, and no body is computed at nature's pair, so at the rule's own matter pair the model does not describe the gravity of the solar system. The findings against nature, listed once in Section 10.3, stand, each the price of one of the model's three virtues, the lattice's frame among them with its mend a hypothesis by name. The two slits and Bell are derived from the rule and read by the engine's gates, no experiment against nature performed; what remains is to compare a body's clicks with nature's. The paper's object is the method, on which the wave equation, the quantum of action and the metric are three readings of one line with nothing continuous beneath; its formulas are examples and Section 10.3 the list of what it does not yet give; we hold the whole to be the least computation we know of that gives the two slits and Bell under the declared credit, with no theorem of minimality; why nature is so, we do not claim to know.

Statements and Declarations

Funding. None. **Competing interests.** None. **Ethics approval and consent.** Not applicable. **Data and code availability.** The law's document, the implementation and the figures' scripts are developed in the public Git repository <https://github.com/Closer24/Universe24> and archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [40], under the MIT licence. **Author contributions.** The author laid down every postulate and decision of the model, directed the work and wrote the paper. **The use of a large language model.**

The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author's direction, for the software, the computations, the derivations and their checks, and the drafting of the text; the model is not an author, and the author reviewed every statement and takes full responsibility for the content. Every figure is drawn by a script from the definitions or from the law's document.

References

- [1] The law of the GameBoard, `docs/ALGEBRA.md` of the archive [40], cited by the name of its section: the GameBoard and the families (the objects, a family's declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count is the record's share, the click is the detector's report, no click names a Node, the algebra of the exclusivity); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, with no waves on the event board, what produces a click is not known and nature's numbers enter the board only as clicks; the hypotheses under their own names; the postulates; the constants; the rows against nature); what is open.
- [2] The engine, `docs/ENGINE.md` of the archive [40]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, `docs/HIGHLIGHTS.md` of the archive [40]: the model's decisions in force, one line each, cited by the line's name.
- [4] K. Zuse, *Rechnender Raum* (Vieweg, Braunschweig, 1969); English translation: *Calculating Space*, MIT Technical Translation AZT-70-164-GEMIT (MIT Project MAC, Cambridge, 1970).
- [5] E. Fredkin and T. Toffoli, Conservative logic, *International Journal of Theoretical Physics* 21, 219 (1982), doi:10.1007/BF01857727.
- [6] N. Margolus, Physics-like models of computation, *Physica D* 10, 81 (1984), doi:10.1016/0167-2789(84)90252-5.
- [7] T. Toffoli and N. Margolus, Invertible cellular automata: a review, *Physica D* 45, 229 (1990), doi:10.1016/0167-2789(90)90185-R.
- [8] S. Wolfram, *A New Kind of Science* (Wolfram Media, Champaign, 2002).
- [9] G. 't Hooft, Duality between a deterministic cellular automaton and a bosonic quantum field theory in 1+1 dimensions, *Foundations of Physics* 43, 597 (2013), doi:10.1007/s10701-013-9702-z.
- [10] G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics*, *Fundamental Theories of Physics* 185 (Springer, Cham, 2016), doi:10.1007/978-3-319-41285-6.
- [11] I. Bialynicki-Birula, Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, *Physical Review D* 49, 6920 (1994), doi:10.1103/PhysRevD.49.6920.
- [12] D. A. Meyer, From quantum cellular automata to quantum lattice gases, *Journal of Statistical Physics* 85, 551 (1996), doi:10.1007/BF02199356.
- [13] P. Arrighi, An overview of quantum cellular automata, *Natural Computing* 18, 885 (2019), doi:10.1007/s11047-019-09762-6.
- [14] T. Farrelly, A review of quantum cellular automata, *Quantum* 4, 368 (2020), doi:10.22331/q-2020-11-30-368.
- [15] G. M. D'Ariano and P. Perinotti, Derivation of the Dirac equation from principles of information processing, *Physical Review A* 90, 062106 (2014), doi:10.1103/PhysRevA.90.062106.
- [16] R. P. Feynman, Simulating physics with computers, *International Journal of Theoretical Physics* 21, 467 (1982), doi:10.1007/BF02650179.
- [17] J. A. Wheeler and R. P. Feynman, Interaction with the absorber as the mechanism of radiation, *Reviews of Modern Physics* 17, 157 (1945), doi:10.1103/RevModPhys.17.157.
- [18] J. G. Cramer, The transactional interpretation of quantum mechanics, *Reviews of Modern Physics* 58, 647 (1986), doi:10.1103/RevModPhys.58.647.
- [19] Y. Aharonov, P. G. Bergmann and J. L. Lebowitz, Time symmetry in the quantum process of measurement, *Physical Review* 134, B1410 (1964), doi:10.1103/PhysRev.134.B1410.
- [20] Y. Aharonov and L. Vaidman, The two-state vector formalism: an updated review, *Lecture Notes in Physics* 734, 399 (Springer, Berlin, 2008), doi:10.1007/978-3-540-73473-4_13.
- [21] H. Price, *Time's Arrow and Archimedes' Point: New Directions for the Physics of Time* (Oxford University Press, New York, 1996).
- [22] K. B. Wharton and N. Argaman, Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics, *Reviews of Modern Physics* 92, 021002 (2020), doi:10.1103/RevModPhys.92.021002.
- [23] N. Harrigan and R. W. Spekkens, Einstein, incompleteness, and the epistemic view of quantum states, *Foundations of Physics* 40, 125 (2010), doi:10.1007/s10701-009-9347-0.
- [24] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics Physique Fizika* 1, 195 (1964), doi:10.1103/PhysicsPhysiqueFizika.1.195.
- [25] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories, *Physical Review Letters* 23, 880 (1969), doi:10.1103/PhysRevLett.23.880.
- [26] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Physical Review Letters* 48, 291 (1982), doi:10.1103/PhysRevLett.48.291.
- [27] P. M. Pearle, Hidden-variable example based upon data rejection, *Physical Review D* 2, 1418 (1970), doi:10.1103/PhysRevD.2.1418.
- [28] P. H. Eberhard, Background level and counter efficiencies required for a loophole-free Einstein-Podolsky-Rosen experiment, *Physical Review A* 47, R747 (1993), doi:10.1103/PhysRevA.47.R747.
- [29] A. Garg and N. D. Mermin, Detector inefficiencies in the Einstein-Podolsky-Rosen experiment, *Physical Review D* 35, 3831 (1987), doi:10.1103/PhysRevD.35.3831.
- [30] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* 526, 682 (2015), doi:10.1038/nature15759.
- [31] M. Giustina et al., Significant-loophole-free test of Bell's theorem with entangled photons, *Physical Review Letters* 115, 250401 (2015), doi:10.1103/PhysRevLett.115.250401.

- [32] L. K. Shalm et al., Strong loophole-free test of local realism, *Physical Review Letters* 115, 250402 (2015), doi:10.1103/PhysRevLett.115.250402.
- [33] T. Regge, General relativity without coordinates, *Il Nuovo Cimento* 19, 558 (1961), doi:10.1007/BF02733251.
- [34] L. Bombelli, J. Lee, D. Meyer and R. D. Sorkin, Space-time as a causal set, *Physical Review Letters* 59, 521 (1987), doi:10.1103/PhysRevLett.59.521.
- [35] C. M. Will, The confrontation between general relativity and experiment, *Living Reviews in Relativity* 17, 4 (2014), doi:10.12942/lrr-2014-4.
- [36] J. Collins, A. Perez, D. Sudarsky, L. Urrutia and H. Vucetich, Lorentz invariance and quantum gravity: an additional fine-tuning problem?, *Physical Review Letters* 93, 191301 (2004), doi:10.1103/PhysRevLett.93.191301.
- [37] D. Mattingly, Modern tests of Lorentz invariance, *Living Reviews in Relativity* 8, 5 (2005), doi:10.12942/lrr-2005-5.
- [38] R. Courant, K. Friedrichs and H. Lewy, Über die partiellen Differenzengleichungen der mathematischen Physik, *Mathematische Annalen* 100, 32 (1928), doi:10.1007/BF01448839.
- [39] H. Yilmaz, New approach to general relativity, *Physical Review* 111, 1417 (1958), doi:10.1103/PhysRev.111.1417.
- [40] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), *Zenodo* (2026), doi:10.5281/zenodo.22738746 (the concept DOI, resolving to the latest version).
- [41] M. Born, Zur Quantenmechanik der Stoßvorgänge, *Zeitschrift für Physik* 37, 863 (1926), doi:10.1007/BF01397477.
- [42] H. Yukawa, On the interaction of elementary particles. I, *Proceedings of the Physico-Mathematical Society of Japan* 17, 48 (1935), doi:10.11429/ppmsj1919.17.0_48.
- [43] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor and INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *The Astrophysical Journal Letters* 848, L13 (2017), doi:10.3847/2041-8213/aa920c.
- [44] G. H. Derrick, Comments on nonlinear wave equations as models for elementary particles, *Journal of Mathematical Physics* 5, 1252 (1964), doi:10.1063/1.1704233.
- [45] S. I. Pohozaev, Eigenfunctions of the equation $\Delta u + \lambda f(u) = 0$, *Soviet Mathematics Doklady* 6, 1408 (1965).
- [46] S. I. Pekar, *Untersuchungen über die Elektronentheorie der Kristalle* (Akademie-Verlag, Berlin, 1954).
- [47] W. H. McCrea and E. A. Milne, Newtonian universes and the curvature of space, *The Quarterly Journal of Mathematics* os-5, 73 (1934), doi:10.1093/qmath/os-5.1.73.
- [48] A. G. Riess et al., Observational evidence from supernovae for an accelerating universe and a cosmological constant, *The Astronomical Journal* 116, 1009 (1998), doi:10.1086/300499.
- [49] S. Perlmutter et al., Measurements of Omega and Lambda from 42 high-redshift supernovae, *The Astrophysical Journal* 517, 565 (1999), doi:10.1086/307221.
- [50] D. Brout et al., The Pantheon+ analysis: cosmological constraints, *The Astrophysical Journal* 938, 110 (2022), doi:10.3847/1538-4357/ac8e04.
- [51] R. V. Pound and G. A. Rebka, Apparent weight of photons, *Physical Review Letters* 4, 337 (1960), doi:10.1103/PhysRevLett.4.337.
- [52] B. Bertotti, L. Iess and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* 425, 374 (2003), doi:10.1038/nature01997.
- [53] I. I. Shapiro, Fourth test of general relativity, *Physical Review Letters* 13, 789 (1964), doi:10.1103/PhysRevLett.13.789.
- [54] S. Coleman and S. L. Glashow, High-energy tests of Lorentz invariance, *Physical Review D* 59, 116008 (1999), doi:10.1103/PhysRevD.59.116008.
- [55] F. W. Stecker, Limiting superluminal electron and neutrino velocities using the 2010 Crab Nebula flare and the IceCube PeV neutrino events, *Astroparticle Physics* 56, 16 (2014), doi:10.1016/j.astropartphys.2014.02.007.
- [56] The Event Horizon Telescope Collaboration, First Sagittarius A* Event Horizon Telescope results. VI. Testing the black hole metric, *The Astrophysical Journal Letters* 930, L17 (2022), doi:10.3847/2041-8213/ac6756.
- [57] The Event Horizon Telescope Collaboration, First M87 Event Horizon Telescope results. I. The shadow of the supermassive black hole, *The Astrophysical Journal Letters* 875, L1 (2019), doi:10.3847/2041-8213/ab0ec7.
- [58] S. Hossenfelder and T. Palmer, Rethinking superdeterminism, *Frontiers in Physics* 8, 139 (2020), doi:10.3389/fphy.2020.00139.
- [59] Z. Cao et al. (LHAASO Collaboration), Ultrahigh-energy photons up to 1.4 petaelectronvolts from 12 gamma-ray Galactic sources, *Nature* 594, 33 (2021), doi:10.1038/s41586-021-03498-z.
- [60] V. A. Acciari et al. (MAGIC Collaboration), Bounds on Lorentz invariance violation from MAGIC observation of GRB 190114C, *Physical Review Letters* 125, 021301 (2020), doi:10.1103/PhysRevLett.125.021301.
- [61] M. Zych, F. Costa, I. Pikovski and C. Brukner, Quantum interferometric visibility as a witness of general relativistic proper time, *Nature Communications* 2, 505 (2011), doi:10.1038/ncomms1498.
- [62] R. C. Tolman, P. Ehrenfest and B. Podolsky, On the gravitational field produced by light, *Physical Review* 37, 602 (1931), doi:10.1103/PhysRev.37.602.
- [63] S. W. Hawking, Particle creation by black holes, *Communications in Mathematical Physics* 43, 199 (1975), doi:10.1007/BF02345020.
- [64] C. W. F. Everitt et al., Gravity Probe B: final results of a space experiment to test general relativity, *Physical Review Letters* 106, 221101 (2011), doi:10.1103/PhysRevLett.106.221101.
- [65] L. Shao and N. Wex, New tests of local Lorentz invariance of gravity with small-eccentricity binary pulsars, *Classical and Quantum Gravity* 29, 215018 (2012), doi:10.1088/0264-9381/29/21/215018.
- [66] F. J. Hasert et al., Observation of neutrino-like interactions without muon or electron in the Gargamelle neutrino experiment, *Physics Letters B* 46, 138 (1973), doi:10.1016/0370-2693(73)90499-1.
- [67] The ALEPH, DELPHI, L3, OPAL and SLD Collaborations, Precision electroweak measurements on the Z resonance, *Physics Reports* 427, 257 (2006), doi:10.1016/j.physrep.2005.12.006.
- [68] C. S. Wu, E. Ambler, R. W. Hayward, D. D. Hoppes and R. P. Hudson, Experimental test of parity conservation in beta decay, *Physical Review* 105, 1413 (1957), doi:10.1103/PhysRev.105.1413.